FINALTERM EXAMINATION
Fall 2009
Calculus & Analytical Geometry-I

Question No: 1 (Marks: 1) - Please choose one

Let $y = \frac{x^2}{2}$. Find average rate of change of y with respect to x over the interval [3, 4]

- ► $\frac{25}{2}$
- ► $\frac{7}{2}$
- ► $\frac{25}{14}$
- ► $\frac{7}{14}$

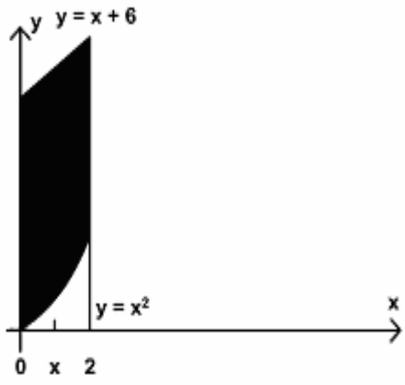
Question No: 2 (Marks: 1) - Please choose one

If $2x - y = -3$ then $\frac{dy}{dx} =$

- ► **2**
- ► -2
- ► 0
- ► -3

Question No: 3 (Marks: 1) - Please choose one

In the following figure, the area bounded on the sides by the lines are **:**



- $x = 0$
- $x = 2$
- $x = 0 \; and \; x = 2$
- $x = 6$

Question No: 4 (Marks: 1) - Please choose one

What is the sum of following series?

$1 + 2 + 3 + 4 + _____ + n$

- $\frac{n+1}{2}$
- $\frac{(n+1)(n+2)}{2}$
- $\frac{n(n+2)}{2}$
- $\frac{n(n+1)}{2}$

Question No: 5 (Marks: 1) - Please choose one

Let f is a smooth function on [0, 3]. What will be the arc length L of the curve y = f(x) from x = 0 to x = 3?

- $L = \int_0^3 \sqrt{1 + [f(x)]^2}\, dy$

$$L=\int_a^b \sqrt{1+[f'(x)]^2}$$

►

$$L=\int_0^3 \sqrt{1+[f'(x)]^2}\,dy$$

►

$$L=\int_0^3 \sqrt{1+[f'(x)]^2}\,dx$$

►

Question No: 6 (Marks: 1) - Please choose one

The PYTHAGORAS theorem describes the relationship between the sides of .....► **Right angle triangle** .....

► **Right angle triangle**
► Isoceleous triangle
► Equilateral triangle

Question No: 7 (Marks: 1) - Please choose one

Which operation can not be applied on the functions?

► Subtraction
► **Cross product**
► Addition
► Composition

Question No: 8 (Marks: 1) - Please choose one

The graph of the equation $y=x^2-4x+5$ will represent

► **Parabola**
► Straight line
► Two straight lines

► Ellipse

Question No: 9 (Marks: 1) - Please choose one

Polynomials are always ..... functions

► **Continuous**
► Discontinuous

Question No: 10 (Marks: 1) - Please choose one

The tan(x) is discontinuous at the points where

- ► Cos(x) =0
- ► Sin(x) =0
- ► **Tan(x) =0**

Question No: 11 (Marks: 1) - Please choose one

A differentiable function must be differentiable on the interval

$(-\infty,\infty)$

- ► $(0,\infty)$
- ► $(-\infty,\infty)$
- ► (a,∞) *where a is any negative* int *eger*

Question No: 12 (Marks: 1) - Please choose one

Let $y=(x^3+2x)^{37}$. Which of the following is correct?

- ► $\frac{dy}{dx}=(37)(x^3+2x)^{36}$
- ► $\frac{dy}{dx}=111x^2(x^3+2x)^{36}$
- ► $\frac{dy}{dx}=(111x^2+74)(x^3+2x)^{36}$
- ► $\frac{dy}{dx}=(111x^2+74)(x^3+2x)^{38}$

Question No: 13 (Marks: 1) - Please choose one

$$\int\frac{3x^2+4x+1}{x^3+2x^2+x-3}\,dx$$

Consider the indefinite integral

Let $t=x^3+2x^2+x-3$

Is the following substitution correct?

$$\int\frac{3x^2+4x+1}{x^3+2x^2+x-3}\,dx=\int\frac{1}{t}\,dt$$

- ► Yes
- ► **No**

Question No: 14 (Marks: 1) - Please choose one

$\log_b ac =$ _____

► $\log_b a + \log_b c$

► $\log_b a - \log_b c$

► $\dfrac{\log_b a}{\log_b c}$

► $(\log_b a)(\log_b c)$

Question No: 15 (Marks: 1) - Please choose one

If a function has an extreme value (either a maximum or a minimum) on an open interval (a,b), then the extreme value occurs at ..... of f

► First point

► Mid point

► **Critical point**

► End point

Question No: 16 (Marks: 1) - Please choose one

The Mean Value Theorem states that "Let function *f* be differentiable on (a,b) and continuous on [a, b], then there exist at least one point c in (a,b) where ..... "

► $f'(c) = \dfrac{f(b) - f(a)}{b - a}$

► $f(c) = \dfrac{f(b) - f(a)}{b - a}$

► $f(c) = \dfrac{f(a) - f(b)}{b - a}$

► $f'(c) = \dfrac{f(a) - f(b)}{b - a}$

Question No: 17 (Marks: 1) - Please choose one

If there is some function *F* such that $\dfrac{d}{dx}[F(x)] = f(x)$ then any function of the form $F(x) + C$ is ---------- of *f*(*x*)

► Derivative
► **Antiderivative**
► Slope
► Maximum value

Question No: 18 (Marks: 1) - Please choose one

$$\sum_{k=1}^{n} f(x^*_k)\,\Delta x_k$$

The sum is known as:

► **Riemann Sum**
► General Sum
► Integral Sum
► Geometric Sum

Question No: 19 (Marks: 1) - Please choose one

$$\int_0^{\frac{\pi}{2}} \cos u\, du$$

If , then which of the following is true?

► 2
► **1**
► 0
► -1

Question No: 20 (Marks: 1) - Please choose one

$$\int_0^{\pi} \sin u\, du$$

If , then which of the following is true?

► 1
► 2
► **0**
► -1

Question No: 21 (Marks: 1) - Please choose one

$$\frac{d}{dx}[F(x)] = f(x)$$

If there is some function *F* such that then antiderivatives of $f(x)$ are $F(x)+C$. What does *C* represents?

► Polynomial
► **Constant**
► Dependent Variable
► Independent Variable

Question No: 22 (Marks: 1) - Please choose one

If *f* and *g* are continues function on an interval [a, b] and $f(x) \geq g(x) \ \ for \ \ a \leq x \leq b$, then area is bounded by the lines parallel to:

► X -axis
► Y-axis
► Both X -axis and Y-axis

Question No: 23 (Marks: 1) - Please choose one

$\int_{1}^{2/3} dx = _____$

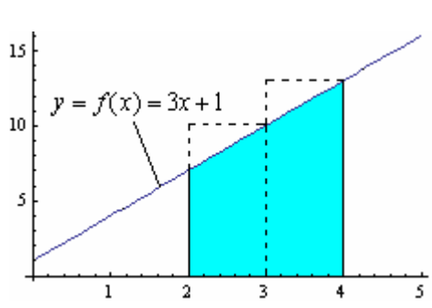
► $\frac{-1}{3}$
► 0
► $\frac{1}{3}$
► $\frac{2}{3}$

Question No: 24 (Marks: 1) - Please choose one

$\int_{0}^{2} x \ dx = _____$

► **2**

► 0
► **2**
► -2
► $\frac{x^2}{2}$

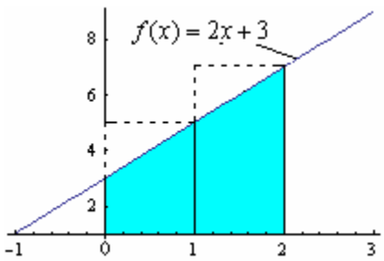


Question No: 25 (Marks: 1) - Please choose one

Which of the following is approximate area of the shaded region by taking x_1^* and x_2^* as left endpoint of equal-length subintervals?

► 17
► 20
► **23**
► 26

Question No: 26 (Marks: 1) - Please choose one



Which of the following is approximate area of the shaded region by taking x_1^* and x_2^* as right endpoint of equal-length subintervals?

► 8
► 10
► 12
► 14

Question No: 27 (Marks: 1) - Please choose one

What is the length of each sub-interval, if the interval [1,3] is divided into *n* sub-intervals of equal length?

► $\frac{1}{n}$
► $\frac{2}{n}$
► $\frac{3}{n}$
► $\frac{4}{n}$

Question No: 28 (Marks: 1) - Please choose one

Evaluate

$$\lim_{x\to 2}\frac{x^2-4}{x-2}=-----$$

► **4**
► 2
► 1
► ∞

Question No: 29 (Marks: 1) - Please choose one

$$\left\{\frac{1}{2^n}\right\}_1^n$$

represents the sequence**:**

$$\frac{-1}{2},\frac{-1}{4},\frac{-1}{8},\cdots$$

►

► $\frac{1}{2},\frac{1}{4},\frac{1}{8},\ldots$

► $0,1,\frac{1}{2},\frac{1}{4},\ldots$

► $0,1,2,3\ldots$

Question No: 30 (Marks: 1) - Please choose one

For a sequence $\{a_n\}$ if the difference between successive terms $a_{n+1}-a_n \le 0$ then the sequence is known as**:**

► Increasing
► Decreasing
► Nondecreasing
► **Nonincreasing**

Question No: 31 (Marks: 1) - Please choose one

For a sequence $\{a_n\}$ if the ratio of successive terms $\frac{a_{n+1}}{a_n}>1$ then the sequence is known as**:**

► **Increasing**
► Decreasing
► Nondecreasing
► Nonincreasing

Question No: 32 (Marks: 1) - Please choose one

If the partial sum of a series is finite then the series will/will be**:**

► Divergent
► **Convergent**
► Give no information

Question No: 33 (Marks: 1) - Please choose one

If the geometric series $a+ar+ar^2+ar^3+\ldots+ar^{k-1}+\ldots \ \ where\ (a\neq 0)$, $|r|<1$ then which of the following is true for the given series?

► **Converges**
► Diverges
► Gives no information

Question No: 34 (Marks: 1) - Please choose one

If $\rho=\lim_{k\to+\infty}\sqrt[k]{u_k}$ where $\rho>1$ then the series $\sum u_k$ with positive terms will /will be.....?

► **Convergent**

► Divergent
► Give no information

Question No: 35 (Marks: 1) - Please choose one

$$\sum_{k=1}^{\infty}(-1)^{k+1}\frac{1}{k}$$

Which of the following is true for the series ?

► Arithmetic Series
► Geometric Series
► **Alternating Harmonic Series**
► Harmonic Series

Question No: 36 (Marks: 1) - Please choose one

.....is the special case of Tylor's theorem.

► **Roll's Theorem**
► Picard's Method
► Integration
► Maclaurin's Theorem

Question No: 37 (Marks: 1) - Please choose one

If f is integrable on a closed interval containing the four points *a*, *b*, *c* and *d* then

$$\int_a^d f(x)dx = _____$$

► $\int_a^b f(x)\,dx+\int_b^c f(x)\,dx+\int_c^d f(x)\,dx$

► $\int_a^b f(x)\,dx+\int_c^d f(x)\,dx$

► $\int_a^c f(x)\,dx+\int_b^d f(x)\,dx$

► $\int_a^d f(x)\,dx$

►

Question No: 38 (Marks: 1) - Please choose one

Suppose f and g are integrable functions on [*a*, *b*] and *c* is a constant, then

$$\int_a^b c\,[f(x)+g(x)]\,dx = _____$$

$$\int_a^b f(cx)dx + \int_a^b g(cx)dx$$

►

$$\int_a^b f(x)\ dx + \int_a^b g(x)]\ dx$$

►

$$c\int_a^b f(x)dx + c\int_a^b g(x)dx$$

►

► 0

Question No: 39 (Marks: 1) - Please choose one

What is the difference between the values of the

$$\int_a^b f(x)dx \quad and \quad \int_a^b f(t)dt$$

integrals ?

► Differ by b-a
► Differ by a-b
► **No difference**
► Differ by b+a

Question No: 40 (Marks: 1) - Please choose one

If $\int_{-1}^{2} f(x)\ dx = 5$ and $\int_{-1}^{2} g(x)\ dx = -3$ then which of the following is value

$$\int_{-1}^{2} [f(x) + 2g(x)]\ dx$$

of ?

► **-1**
► -8
► 2
► 11

Question No: 41 (Marks: 2)

Express the sum $\frac{1}{1} + \frac{1}{8} + \frac{1}{27} + ... + \frac{1}{1000}$ in sigma notation.

$$\sum_{n=1}^{10} (1/n^3)$$

Question No: 42 (Marks: 2)

Only write down the Maclaurin series for e^x

Question No: 43 (Marks: 2)

Evaluate the following integral:

$$\int_1^4 \sqrt{x}\ dx$$

$$\int_1^4 \sqrt{x}\, dx$$

$$= \int_1^4 \sqrt{x}.1\, dx$$

$$= x\sqrt{x} + \int_1^4 1/\sqrt{x}.1\, dx$$

Question No: 44 (Marks: 3)

Evaluate the following sum:

$$\sum_{k=1}^{6} (k^2 - 5)$$

$$= -4 - 1 + 4 + 11 + 20 + 31 = 61$$

Question No: 45 (Marks: 3)

Find a definite integral indicating the area enclosed by the curves $y = x^2 \ , \ x > 0$ and bounded on the sides by the lines $y = 1$ and $y = 4$. But do not evaluate the integral.

Question No: 46 (Marks: 3)

$$a_n = \left\{ \frac{3}{n^2} \right\}_{n=5}^{\infty}$$

Determine whether the following sequence is strictly monotone or not.If your answer is yes or no, then give reason .

Yes the sequence is strictly monotone because the denominator is increasing

Question No: 47 (Marks: 5)

The region bounded by the *y*-axis, the graph of the equation $x = y^{\frac{3}{2}}$ and the line *y* = 2 is revolved about *y*-axis. Find the volume of the resulting solid.

Question No: 48 (Marks: 5)

Compute the following sum:

$$\sum_{i=1}^{n} \left(4i^2 - i\right) = \left(4(1)^2 - 1\right) + \left(4(2)^2 - 2\right) + \left(4(3)^2 - 3\right) + \left(4(4)^2 - 4\right) \ldots\ldots\ldots\ldots$$

$$= 3 + 14 + 33 + 60 \ldots\ldots\ldots\ldots\ldots$$

Question No: 49 (Marks: 5)

Use L'Hopital's rule to evaluate the limit

$$\lim_{x\to\frac{\pi}{2}} \frac{1-\sin x}{1+\cos 2x}$$

$$\lim_{x\to\frac{\pi}{2}} (1-\sin x)=0 \lim_{x\to\frac{\pi}{2}} (1+\cos 2x)=0$$

$$=0/0$$

So by L'Hopital's rule

$$\lim_{x\to\frac{\pi}{2}} \frac{1-\sin x}{1+\cos 2x}$$

$$=\lim_{x\to\frac{\pi}{2}} \frac{d/dx(1-\sin x)}{d/dx(1+\cos 2x)}$$

$$=\lim_{x\to\frac{\pi}{2}} \frac{-\cos x}{-2\sin 2x}=\frac{\cos\frac{\pi}{2}}{2\sin\pi}=0$$

Question No: 50 (Marks: 10)

$$\sum_{1}^{\infty} \frac{2^n}{n(n+2)}$$

Use the Ratio test to determine whether the series converges or diverges.

$$p=\lim_{x\to\infty} \frac{u_{k+1}}{u_k}=$$

FINALTERM EXAMINATION 2009
(Session - 2)

Calculus & Analytical Geometry-I

Question No: 1 (Marks: 1) - Please choose one

According to Power-Rule of differentiation, if $f(x) = x^n$ where n is a real number, then

$\frac{d}{dx}[x^n] =$

- ▶ x^{n-1}
- ▶ $n\,x^{n-1}$
- ▶ $n\,x^{n+1}$
- ▶ $(n-1)x^{n+1}$

Question No: 2 (Marks: 1) - Please choose one

If a function g is differentiable at a point x and a function f is differentiable at a point g(x), then the _____ is differentiable at point x .

- ▶ Composition (f o g)
- ▶ Quotient (f / g)
- ▶ Product (f . g)
- ▶ Sum (f + g)

Question No: 3 (Marks: 1) - Please choose one

$y = f\left(g\left(h(x)\right)\right)$

If

$u = g\left(h(x)\right)$

$v = h(x)$ $\frac{dy}{dx} =$ _____

then

- ▶ $\frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$
- ▶ $\frac{dy}{du}\frac{du}{dv}\frac{dv}{dx}$
- ▶ $\frac{dv}{du} \cdot \frac{du}{dv} \cdot \frac{dy}{dx}$

Question No: 4 (Marks: 1) - Please choose one

If a function f is ..... on a closed interval [a,b] , then f has both maximum and minimum value on [a,b].

- ▶ Continuous
- ▶ Discontinuous
- ▶ None of these

Question No: 5 (Marks: 1) - Please choose one

$$\int_a^x \frac{t^2}{2} dt$$

The expression , represents a function of :

- ▶ t
- ▶ a
- ▶ Both x and a
- ▶ x

Question No: 6 (Marks: 1) - Please choose one

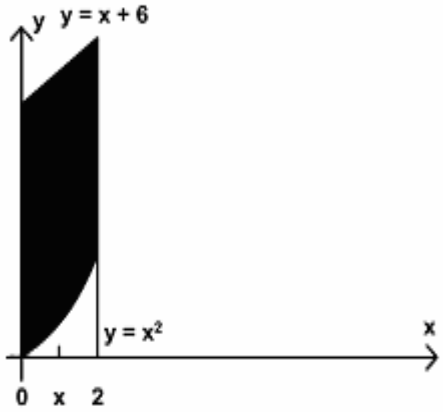
$\int cf(x)dx =$ _____

if c is a constant

- 0
- c
- $\int f(cx)dx$
- $c\int f(x)dx$

Question No: 7 (Marks: 1) - Please choose one

In the following figure, the area enclosed is bounded below by **:**



- $y = x + 6$
- $y = x^2$
- $x = 2$
- $x = 0$

Question No: 8 (Marks: 1) - Please choose one

At what points the two curves: $y = x^2 \ and \ y = x + 6$ intersect ?

- $x = 0 \ and \ x = 2$
- $x = 0 \ and \ x = 3$
- $x = 2 \ and \ x = 3$
- $x = -2 \ and \ x = 3$

Question No: 9 (Marks: 1) - Please choose one

What is the sum of following series?

$1+2+3+4+_____+n$

- $\frac{n+1}{2}$
- $\frac{(n+1)(n+2)}{2}$
- $\frac{n(n+2)}{2}$
- $\frac{n(n+1)}{2}$

Question No: 10 (Marks: 1) - Please choose one

$If\ b>0\ then\ \frac{d}{dx}[b^x] = _____$

- 0
- xb^{x-1}
- $\ln b$
- $b^x \ln b$

Question No: 11 (Marks: 1) - Please choose one

Let S be a solid bounded by two parallel planes perpendicular to the x-axis at x = a and x = b .If , for each x in [a,b], the cross-section area of S perpendicular to the x-axis is A(x) ,then what is the volume of the solid?

- $V=\int_a^b A(y)dx$
- $V=\int_a^b A(x)dx$

$$V = \int_{0}^{A(x)} [b-a]dx$$

▶

$$V = \int_{0}^{A(x)} [b+a]dx$$

▶

Question No: 12 (Marks: 1) - Please choose one

Let the solid generated by the region enclosed between $y = \sqrt{x}$; $x = 1, x = 4$ and the x-axis is revolved about the y-axis. Which of the following equation gives the volumes of a solid by cylindrical shells?

$$V = \int_{1}^{4} 2\pi x\sqrt{x}dx$$

▶

$$V = \int_{1}^{4} 2x\sqrt{x}dx$$

▶

$$V = \int_{0}^{4} 2x\sqrt{x}dx$$

▶

$$V = \int_{-4}^{4} 2x\sqrt{x}dx$$

▶

Question No: 13 (Marks: 1) - Please choose one

If slope m of a none vertical line is m =1 then the angle of inclination of the line is

$$\frac{\pi}{4}$$

▶

► $\frac{\pi}{2}$

► $\frac{\pi}{5}$

Question No: 14 (Marks: 1) - Please choose one

The PYTHAGORAS theorem describes the relationship between the sides of .....

► Right angle triangle
► Isoceleous triangle
► Equilateral triangle

Question No: 15 (Marks: 1) - Please choose one

If a quantity y depends on another quantity x in such a way that each value of x determines exactly one value of y , we say that y is ..... of x

► relation
► function
► not a function
► not a relation

Question No: 16 (Marks: 1) - Please choose one

The graph of the equation $y = x^2 - 4x + 5$ will represent

► Parabola
► Straight line
► Two straight lines

► Ellipse

Question No: 17 (Marks: 1) - Please choose one

The $\lim_{x \to a} f(x) = ...where\ f(x) = k$ (k is a constant)

► k+2
► k+1
► k

Question No: 18 (Marks: 1) - Please choose one

$$\int \frac{1}{x^2+2}\, dx$$

Consider the indefinite integral

Let $t = x^2 + 2$

Is the following substitution correct?

$\int \frac{1}{x^2+2}\, dx = \int \frac{1}{t}\, dt$

► Yes
► No

Question No: 19 (Marks: 1) - Please choose one

$\log_b \frac{1}{t} =$ _____

► $\log_b t$
► $1 - \log_b t$
► $1 + \log_b t$
► $-\log_b t$

Question No: 20 (Marks: 1) - Please choose one

How the series $1-3+5-7+9-11$ can be expressed in sigma notation?

► $\sum_{k=0}^{k=5} (-1)^k (2k+1)$

$$\sum_{k=1}^{k=5}(-1)^k(2k+1)$$

►

$$\sum_{k=1}^{k=5}(2k+1)$$

►

$$\sum_{k=1}^{k=5}(2k+1)$$

►

Question No: 21 (Marks: 1) - Please choose one

$$\sum_{k=1}^{n} f(x^*_k)\Delta x_k$$

The sum is known as:

► Riemann Sum
► General Sum
► Integral Sum
► Geometric Sum

Question No: 22 (Marks: 1) - Please choose one

$$\sum_{k=1}^{n} f(x^*_k)\Delta x_k$$

What does 'n' represent in the Riemann Sum ?

► No. of Circles
► No. of Subintervals
► No. of Loops
► No. of Squares

Question No: 23 (Marks: 1) - Please choose one

If *w* and *v* are continuous functions of *y* on an interval [c, d] and $w(y) \geq v(y) \ for \ c \leq y \leq d$, then area is bounded by the lines parallel to: **:**

► Y-axis
► X-axis
► Both X-axis and Y-axis

Question No: 24 (Marks: 1) - Please choose one

How the series $2(1)+2(2)+2(3)+2(4)+2(5)$ can be expressed in sigma notation?

- $\sum_{k=0}^{5} 2k^2$
- $\sum_{k=1}^{5} 2k^2$
- $\sum_{k=0}^{5} 2k$
- $\sum_{k=1}^{5} 2k$

Question No: 25 (Marks: 1) - Please choose one

$\sum_{k=1}^{n} \frac{k^3}{2} =$ _____

- $\frac{n(n+1)}{4}$
- $\frac{[n(n+1)]^2}{8}$
- $\frac{n(n+1)(2n+1)}{12}$
- $\frac{(n+1)(2n+1)}{6}$

Question No: 26 (Marks: 1) - Please choose one

If $a_1 < a_2 < a_3 < ..... < a_n < ....$, then a sequence $\{a_n\}$ is.....

- Nondecreasing
- Decreasing

► Increasing
► Nonincreasing

Question No: 27 (Marks: 1) - Please choose one

If $a_1 \geq a_2 \geq a_3 \geq ..... \geq a_n \geq ....$, then a sequence $\{a_n\}$ is .....

► Increasing
► Nondecreasing
► Decreasing
► Nonincreasing

Question No: 28 (Marks: 1) - Please choose one

If the difference between successive terms $a_{n+1} - a_n > 0$ then the sequence $\{a_n\}$ is known as**:**

► Increasing
► Decreasing
► Nondecreasing
► Nonincreasing

Question No: 29 (Marks: 1) - Please choose one

For a sequence $\{a_n\}$ if the ratio of successive terms $\frac{a_{n+1}}{a_n} > 1$ then the sequence is known as**:**

► Increasing
► Decreasing
► Nondecreasing
► Nonincreasing

Question No: 30 (Marks: 1) - Please choose one

For a sequence $\{a_n\}$ if the ratio of successive terms $\frac{a_{n+1}}{a_n} \geq 1$ then the sequence is known as **:**

► Increasing
► Decreasing
► Nondecreasing

► Nonincreasing

Question No: 31 (Marks: 1) - Please choose one

If $f(n) = a_n$ is the nth term of the sequence and $f(n)$ is differentiable and $f'(n) > 0$ then the sequence will be**:**

► Increasing
► Decreasing
► Nondecreasing
► Nonincreasing

Question No: 32 (Marks: 1) - Please choose one

If the geometric series $a + ar + ar^2 + ar^3 + ... + ar^{k-1} + ... \ \ where\ (a \neq 0)$, $|r| < 1$ then which of the following is true for the given series?

► Converges
► Diverges
► Gives no information

Question No: 33 (Marks: 1) - Please choose one

If the geometric series $a + ar + ar^2 + ar^3 + ... + ar^{k-1} + ... \ \ where\ (a \neq 0)$, $|r| \geq 1$ then which of the following is true for the given series?

► Converges
► Diverges
► Gives no information

Question No: 34 (Marks: 1) - Please choose one

If the series $\sum_{k=1}^{\infty} u_k$ converges but the series $\sum_{k=1}^{\infty} |u_k|$ does not converge , then $\sum_{k=1}^{\infty} u_k$ will _____

► Converge absolutely
► Diverge
► Converge conditionally

Question No: 35 (Marks: 1) - Please choose one

Let $\sum u_k$ be a series with nonzero terms and suppose that

$$\rho = \lim_{k\to\infty} \frac{|u_{k+1}|}{|u_k|} > 1$$

then which of the following is true?

► The series $\sum |u_k|$ converges

► The series $\sum |u_k|$ diverges

► No conclusion can be drawn.

Question No: 36 (Marks: 1) - Please choose one

Suppose f and g are integrable functions on [*a*, *b*] and *c* is a constant, then

$\int_a^b c\,[f(x)+g(x)]\,dx =$ _____

► $\int_a^b f(cx)dx + \int_a^b g(cx)dx$

► $\int_a^b f(x)\,dx + \int_a^b g(x)]\,dx$

► $c\int_a^b f(x)dx + c\int_a^b g(x)dx$

►

► 0

Question No: 37 (Marks: 1) - Please choose one

Which of the following is surface area S generated by revolving the curve *y = f(x)* between *x = 0* and *x = 2* about the *x*-axis?

$$S = \int_0^2 2\pi f(x)\sqrt{1+[f(x)]}dx$$

▶

$$S = \int_0^2 2\pi f(x)\sqrt{1+[f'(x)]^2}dx$$

▶

$$S = \int_0^2 2\pi f(x)\sqrt{1+[f'(x)]}dx$$

▶

$$S = \int_0^2 2\sqrt{1+[f'(x)]}dx$$

▶

Question No: 38 (Marks: 1) - Please choose one

Which of the following is area of the surface generated by revolving the curve $y = 4\sqrt{x}\ ;\ 1 \le x \le 4$ about the *x*-axis?

$$\int_1^4 2\pi\left(4\sqrt{x}\right)\sqrt{1+\left[\left(4\sqrt{x}\right)\right]^2}\,dx$$

▶

$$\int_1^4 2\pi\left(4\sqrt{x}\right)\sqrt{1+\left[\left(4\sqrt{x}\right)'\right]}\,dx$$

▶

$$\int_1^4 2\pi+\sqrt{1+\left[\left(4\sqrt{x}\right)'\right]^2}\,dx$$

▶

$$\int_1^4 2\pi\left(4\sqrt{x}\right)\sqrt{1+\left[\left(4\sqrt{x}\right)'\right]^2}\,dx$$

▶

Question No: 39 (Marks: 1) - Please choose one

Which of the following is the work done *W* if an object moves in the positive direction along a coordinate line while subject to a force *F(x)* in the direction of motion over an interval [0,3]?

$$W = \int_{2}^{3} 3xdx$$

▶

$$W = \int_{0}^{3} 3xdx$$

▶

$$W = \int_{0}^{3} F(x)dx$$

▶

$$W = \int_{3}^{0} F(x)dx$$

▶

Question No: 40 (Marks: 1) - Please choose one

$\int_{1}^{0} f(x)\,dx = 2$ $\int_{0}^{5} f(x)\,dx = 1$

If and then which of the following is value of $\int_{1}^{5} f(x)\,dx$?

▶ -3
▶ -1
▶ 1
▶ 3

Question No: 41 (Marks: 2)

Derivative of a function is $\frac{1}{2\sqrt{x}}$. Find the original function.

Question No: 42 (Marks: 2)

If $\sum u_k$ is a series with positive terms and $\rho = \lim_{k \to +\infty} \sqrt[k]{u_k}$; then write the three cases for the series to be convergent, divergent or none.

Question No: 43 (Marks: 2)

Evaluate the following integral:

$$\int_1^3 \frac{1}{x^2}\,dx$$

Question No: 44 (Marks: 3)

Use the first fundamental theorem of calculus to evaluate the definite integral:

$$\int_0^2 f(x)\,dx \qquad where \qquad f(x)=\begin{cases} x^2 & ; \quad 0 \le x < 1 \\ x^3 & ; \quad 1 \le x \le 2 \end{cases}$$

Question No: 45 (Marks: 3)

$$\sum_{k=2}^{\infty}(-1)^{k-1}\frac{2^{k-1}}{(k-1)!}$$

Show that the series converges absolutely.

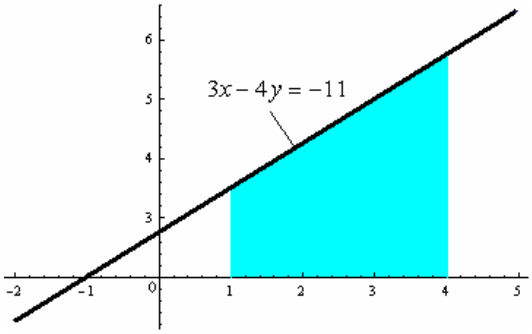
Question No: 46 (Marks: 3)

Express the following definite integral as limit of Riemann Sum. (Do not evaluate the integral)

$$\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} Cos\,x\,dx$$

Question No: 47 (Marks: 5)

Express area of the shaded region as a definite integral.



Question No: 48 (Marks: 5)

How much work is required to wind the chain onto the pulley if a 100-ft length of steel chain weighting
15 lb/ ft. is dangling from a pulley?

Question No: 49 (Marks: 5)

Evaluate the following integral:

$$\int_1^2 \frac{x^2+\sqrt{x}}{x^2}\,dx$$

Question No: 50 (Marks: 10)

Use L'Hopital's Rule to evaluate

$$\lim_{x\to 0}(1+\sin x)^{\cot x}$$

FINALTERM EXAMINATION
Fall 2009
Calculus & Analytical Geometry-I

Question No: 1 (Marks: 1) - Please choose one

Let $f(x)$ is a function such that as x approaches a real number a, either from left or right-hand-side, the function values increases or decreases unboundedly then

$\lim_{x \to a} f(x)$

► Exist
► Does not exist

Question No: 2 (Marks: 1) - Please choose one

$\frac{d(\sec x)}{dx} =$

► $(\sec x)(\tan x)$
► $(\sec x)(\tan x)$
► $(co\sec x)(co\,t\,x)$
► $(co\sec x)(\tan x)$

Question No: 3 (Marks: 1) - Please choose one

Consider a function $h(x)$ and a constant c then

$\frac{d}{dx}\left((c)\{h(x)\}\right) =$ _____

- 0
- $\frac{d}{dx}\left(h(x)\right)$
- $\frac{d}{dx}\left(h(cx)\right)$
- $c\frac{d}{dx}\left(h(x)\right)$

Question No: 4 (Marks: 1) - Please choose one

If f is continuous function such that $\lim_{x\to-\infty} f(x) = +\infty$ *and* $\lim_{x\to+\infty} f(x) = +\infty$ then f has _____ on $(-\infty,+\infty)$

- maximum value but no minimum
- minimum value but no maximum
- both maximum and minimum value

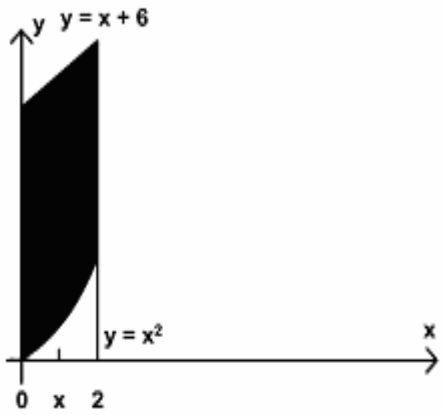
Question No: 5 (Marks: 1) - Please choose one

Sigma notation is represented by which of the following Greek letter?

- χ
- η
- Σ
- ψ

Question No: 6 (Marks: 1) - Please choose one

In the following figure, the area enclosed is bounded below by **:**



- $y = x + 6$
- $y = x^2$
- $x = 2$
- $x = 0$

Question No: 7 (Marks: 1) - Please choose one

At what points the two curves: $y = x^2 \ and \ y = x + 6$ intersect ?

- $x = 0 \ and \ x = 2$
- $x = 0 \ and \ x = 3$
- $x = 2 \ and \ x = 3$
- $x = -2 \ and \ x = 3$

Question No: 8 (Marks: 1) - Please choose one

Let the solid generated by the region enclosed between $y = \sqrt{x} \quad ; \quad x = 1, x = 4$ and the x-axis is revolved about the y-axis. Which of the following equation gives the volumes of a solid by cylindrical shells?

- $V = \int_1^4 2\pi x\sqrt{x}dx$

$$V = \int_{1}^{4} 2x\sqrt{x}dx$$

▶

$$V = \int_{0}^{4} 2x\sqrt{x}dx$$

▶

$$V = \int_{-4}^{4} 2x\sqrt{x}dx$$

▶

Question No: 9 (Marks: 1) - Please choose one

Let f is a smooth curve on the interval [a, b]. What is the arc length L of the curve f(x) defined over the interval [a, b]?

$$L = \lim_{\max \Delta x \to 0} \sum_{k=1}^{n} \sqrt{1 + (f'(x^*_k))}$$

▶

$$L = \sum_{k=1}^{n} \sqrt{1 + (f'(x^*_k))}\Delta x_k$$

▶

$$L = \lim_{\max \Delta x \to 0} \sum_{k=1}^{n} \sqrt{1 + (f'(x^*_k))^2}\Delta x_k$$

▶

$$L = \sum_{k=1}^{n} \sqrt{1 + (f(x^*_k))}\Delta x$$

▶

Question No: 10 (Marks: 1) - Please choose one

For a graph to be symmetric about y-axis means, for each point (x,y) on the graph,the point ---------- is also on the graph

▶ (x , -y)

▶ (-x , y)

► (-x , -y)

Question No: 11 (Marks: 1) - Please choose one

The graph $x = y^2$ is symmetric about ---------axis

► X-axis
► Y-axis
► Origin

Question No: 12 (Marks: 1) - Please choose one

If a quantity y depends on another quantity x in such a way that each value of x determines exactly one value of y , we say that y is ..... of x

► relation
► function
► not a function
► not a relation

Question No: 13 (Marks: 1) - Please choose one

Domain of the function $y = \frac{(x^2-4)}{(x-2)}$ is

► $(-\infty, 2) U (2, +\infty)$
► $(-\infty, 2)$
► $(-\infty, 0)$

Question No: 14 (Marks: 1) - Please choose one

Tan(x) is continuous every where except at points

► $\pm\frac{k\pi}{2}(k = 1, 3, 5, ...)$
► $\pm\frac{k\pi}{2}(k = 2, 4, 6, ...)$

$\pm\frac{k\pi}{2}(k=1,2,3,4,5,6,...)$

▶

Question No: 15 (Marks: 1) - Please choose one

$\lim_{x\to 0}\frac{\sin x}{x}$

= ----------

▶ -1
▶ 2
▶ 0
▶ 1

Question No: 16 (Marks: 1) - Please choose one

How the series $1-3+5-7+9-11$ can be expressed in sigma notation?

$\sum_{k=0}^{k=5}(-1)^k(2k+1)$

▶

$\sum_{k=1}^{k=5}(-1)^k(2k+1)$

▶

$\sum_{k=1}^{k=5}(2k+1)$

▶

$\sum_{k=1}^{k=5}(2k+1)$

▶

Question No: 17 (Marks: 1) - Please choose one

Let the region bounded by the curve $y=\sqrt[3]{x}$, the x-axis, and the line $x=3$ is revolved about the y-axis to generate a solid. Which of the following equation gives the volume of a solid by cylindrical shells?

$$V = \int_0^3 x^{\frac{3}{2}}\, dx$$

▶

$$V = 2\pi \int_0^3 \sqrt{x}\, dx$$

▶

$$V = \int_0^3 2\pi x \sqrt[3]{x}\ dx$$

▶

$$V = \int_0^3 x \sqrt[3]{x}\ dx$$

▶

Question No: 18 (Marks: 1) - Please choose one

$$y = \frac{2\sqrt{2}}{3} x^{\frac{3}{2}}\ ;\ 0 \le x \le 2$$

Let then which of the following is the length of the curve?

$$L = \int_0^2 \sqrt{\left[\frac{d}{dx}\left(\frac{2\sqrt{2}}{3} x^{\frac{3}{2}}\right)\right]^2}\, dx$$

▶

$$L = \int \sqrt{1 + \left[\frac{d}{dx}\left(\frac{2\sqrt{2}}{3} x^{\frac{3}{2}}\right)\right]^2}\, dx$$

▶

$$L = \int_0^2 \sqrt{1 + \left[\frac{d}{dx}\left(\frac{2\sqrt{2}}{3} x^{\frac{3}{2}}\right)\right]^2}\, dx$$

▶

$$L = \int_0^2 \sqrt{1 + \left[\frac{d}{dx}\left(\frac{2\sqrt{2}}{3} x^{\frac{3}{2}}\right)\right]}\, dx$$

▶

Question No: 19 (Marks: 1) - Please choose one

$\frac{2}{3}$

is known as

- An even number
- Irrational Number
- A natural Number
- Rational Number

Question No: 20 (Marks: 1) - Please choose one

For a function *f*, let $f'(x_n) = 0 \quad for\ some\ n$.

Does the Newton's Method works for approximating the solution of $f(x) = 0$?

- Yes
- No

Question No: 21 (Marks: 1) - Please choose one

The Mean Value Theorem states that "Let function *f* be differentiable on (a,b) and continuous on [a, b], then there exist at least one point c in (a,b) where ……. "

- $f'(c) = \frac{f(b) - f(a)}{b - a}$
- $f(c) = \frac{f(b) - f(a)}{b - a}$
- $f(c) = \frac{f(a) - f(b)}{b - a}$
- $f'(c) = \frac{f(a) - f(b)}{b - a}$

Question No: 22 (Marks: 1) - Please choose one

$$\frac{d}{dx}[F(x)] = f(x)$$

If there is some function *F* such that then any function of the form $F(x) + C$ is ---------- of *f*(*x*)

► Derivative
► Antiderivative
► Slope
► Maximum value

Question No: 23 (Marks: 1) - Please choose one

If *f* and *g* are continues function on an interval [a, b]
and $f(x) \geq g(x) \ for \ \ a \leq x \leq b$, then area is bounded by the lines parallel to:

► X -axis
► Y-axis
► Both X -axis and Y-axis

Question No: 24 (Marks: 1) - Please choose one

What is the sum of following series?

$1^3 + 2^3 + 3^3 + 4^3 + _____ + n^3$

► $\frac{n(2n)(2n+1)}{6}$

► $\frac{(n+1)(n+2)}{2}$

► $\left[\frac{n(n+2)}{2}\right]^2$

► $\left[\frac{n(n+1)}{2}\right]^2$

Question No: 25 (Marks: 1) - Please choose one

$\frac{5}{7}\times 1^2 + \frac{5}{7}\times 2^2 + \frac{5}{7}\times 3^2 + \frac{5}{7}\times 4^2 .... + \frac{5}{7}\times n^2 =$ _____

► $\frac{5n(n+1)(2n+1)}{42}$

$\frac{5n(n+1)}{14}$

► $\frac{5n^2(n+1)^2}{14}$

► $\frac{5(n+1)(2n+1)}{42}$

►

Question No: 26 (Marks: 1) - Please choose one

$\int_a^a f(x)dx =$ _____

If point a is in the domain of function f, then

► $f'(x)$

► $f(x)$

► 0

► 1

Question No: 27 (Marks: 1) - Please choose one

If $a_1 > a_2 > a_3 > ..... > a_n > ....$, then a sequence $\{a_n\}$ is .....

► Increasing

► Nondecreasing

► Decreasing

► Nonincreasing

Question No: 28 (Marks: 1) - Please choose one

For a sequence $\{a_n\}$ if the difference between successive terms $a_{n+1} - a_n \leq 0$ then the sequence is known as**:**

► Increasing

► Decreasing

► Nondecreasing

► Nonincreasing

Question No: 29 (Marks: 1) - Please choose one

For a sequence $\{a_n\}$ if the ratio of successive terms $\frac{a_{n+1}}{a_n} < 1$ then the sequence is known as**:**

► Increasing
► Decreasing
► Nondecreasing
► Nonincreasing

Question No: 30 (Marks: 1) - Please choose one

For a sequence $\{a_n\}$ if the ratio of successive terms $\frac{a_{n+1}}{a_n} \geq 1$ then the sequence is known as **:**

► Increasing
► Decreasing
► Nondecreasing
► Nonincreasing

Question No: 31 (Marks: 1) - Please choose one

$$a_n = \left\{\frac{1}{n}\right\}_{n=1}^{\infty}$$

Which of the following option is true for the sequence ?

► Increasing
► Decreasing
► Nonincreasing
► Nondecreasing

Question No: 32 (Marks: 1) - Please choose one

If the partial sum of a series is finite then the series will/will be**:**

► Divergent
► Convergent
► Give no information

Question No: 33 (Marks: 1) - Please choose one

If the geometric series $a + ar + ar^2 + ar^3 + ... + ar^{k-1} + ... \ \ where\ (a \neq 0)$, $|r| < 1$ then which of the following is true for the given series?

- ► Converges
- ► Diverges
- ► Gives no information

Question No: 34 (Marks: 1) - Please choose one

If $\rho = \lim_{k \to +\infty} \frac{u_{k+1}}{u_k}$ where $\rho > 1$ then the series $\sum u_k$ with positive terms will /will be.....?

- ► Convergent
- ► Divergent
- ► Give no information

Question No: 35 (Marks: 1) - Please choose one

If $\rho = \lim_{k \to +\infty} \sqrt[k]{u_k}$ where $\rho > 1$ then the series $\sum u_k$ with positive terms will /will be.....?

- ► Convergent
- ► Divergent
- ► Give no information

Question No: 36 (Marks: 1) - Please choose one

In alternating series test, which one of the following condition must be satisfied?

- ► $\lim_{k \to \infty} a_k = 1$
- ► $a_1 > a_2 > a_3 ..... > a_k > ....$
- ► $a_1 \leq a_2 \leq a_3 ..... \leq a_k \leq ....$

Question No: 37 (Marks: 1) - Please choose one

A series of the form $\sum_{k=1}^{\infty} (-1)^n a_k$ is called _____.

► Alternating series

► Geometric series

► Arithmetic series

► Harmonic series

Question No: 38 (Marks: 1) - Please choose one

Which of the following is the Maclaurin series for e^x ?

► $1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\ldots+\frac{x^k}{k!}+\ldots$

► $x+\frac{x^2}{2!}+\frac{x^3}{3!}+\ldots+\frac{x^k}{k!}+\ldots$

► $1+x+\frac{x^3}{3!}+\ldots+\frac{x^k}{k!}+\ldots$

► $1-x+\frac{x^3}{3!}-\ldots-\frac{x^k}{k!}-\ldots$

Question No: 39 (Marks: 1) - Please choose one

Which of the following is the work done *W* if an object moves in the positive direction along a coordinate line while subject to a force *F(x)* in the direction of motion over an interval [0,3]?

► $W=\int_2^3 3xdx$

► $W=\int_0^3 3xdx$

► $W=\int_0^3 F(x)dx$

$$W = \int_{3}^{0} F(x)dx$$

►

Question No: 40 (Marks: 1) - Please choose one

Which of the following is the spring constant *k* if a spring whose natural length is 2*m* exerts a force of 3*N* when stretched 1*m* beyond its natural length?

► 3 *x*
► 3 *N/m*
► 2 *m*
► 3 *m/N*

Question No: 41 (Marks: 2)

Evaluate the following integral by substitution method.

$$\int x\,(2x^2+1)^{\frac{2}{3}}\,dx$$

Question No: 42 (Marks: 2)

Find the limits of the integral indicating the area bounded by the curves $y = x^2 \text{ and } y = x + 6$.

Sol,

Question No: 43 (Marks: 2)

What will be the amount of work done if an object moves 7*m* in the direction of a force of 70*N*?

Question No: 44 (Marks: 3)

Evaluate the following integral:

$$\int \frac{5 - 6\sin^2 x}{\sin^2 x}\,dx$$

Question No: 45 (Marks: 3)

Find a definite integral indicating the area of the surface generated by revolving the curve $y = \sqrt[3]{3x} \;;\; 0 \le y \le 4$ about the $x-$ axis. But do not evaluate the integral.

Question No: 46 (Marks: 3)

Find the spring constant '*k*'; if a force of 10*N* is required to stretch a spring from its natural length of 4.8*m* to a length of 6.8*m*?

Question No: 47 (Marks: 5)

Let $\frac{d}{dx}[f(x)] = 12x^2 - 6x + 1$. Find $f(x)$

Sol,

Question No: 48 (Marks: 5)

Use the cylindrical shell to find the volume of the solid generated when the region enclosed by the curve $y = x^3 , x = 1, y = 0$ is revolved about the *y*-axis.

Question No: 49 (Marks: 5)

Determine whether the sequence $\{a_n\}$ converges or diverges; if it converges then find its limit;

$$a_n = \frac{3n^4 + 1}{4n^2 - 1}$$

where

Question No: 50 (Marks: 10)

Find the area of the region that is enclosed by the curves $y = x^2$ and $y = \sqrt{x}$ $x = \frac{1}{4}$ and $x = 1$ between .

FINALTERM EXAMINATION
Fall 2008
(Session - 1)

Calculus & Analytical Geometry-I

Question No: 1 (Marks: 1) - Please choose one

If $y = f(x)$ then the average rate of change of y with respect to x over the interval $[x_0, x_1]$ is the ..... Joining the points $(x_0, f(x_0)) and (x_1, f(x_1))$ on the graph of f

- ► Slope of the secant line
- ► Slope of tangent line
- ► Secant line
- ► none of these

Question No: 2 (Marks: 1) - Please choose one

Natural domain of $\frac{(x^2-4)}{(x-2)}$ is

- ► $(-\infty, 2) U (2, +\infty)$
- ► $(-\infty, 2)$
- ► $(-\infty, 0)$
- ► None of these

Question No: 3 (Marks: 1) - Please choose one

The equation $(x+4)^2+(y-1)^2=6$ represents a circle having center at ..... and radius .....

- ► $(-4,1),\sqrt{6}$
- ► $(-4,1),6$
- ► $(-4,-1),\sqrt{6}$
- ► None of these

Question No: 4 (Marks: 1) - Please choose one

The series $\sum u_k$ be a series with positive terms and suppose that $\rho=\lim_{k\to\infty}\sqrt[k]{u_k}=\lim_{k\to\infty}(u_k)^{\frac{1}{k}}$ if $\rho>1$ then the series

- ► Converges
- ► **Diverges**
- ► May converge or diverge
- ► None of these

Question No: 5 (Marks: 1) - Please choose one

The series $\sum u_k \text{ and } \sum v_k$ are convergent series then $(\sum u_k+\sum v_k) \text{ and } (\sum u_k-\sum v_k)$ will be .....and.....

- ► Convergent, convergent
- ► Divergent, divergent
- ► **Convergent, divergent**
- ► Divergent, convergent

Question No: 6 (Marks: 1) - Please choose one

The notation $\{\frac{1}{2^n}\}_{-1}^{n}$ represents the sequence

► $2,1,\frac{1}{2},\frac{1}{4},...$

► $0,1,2,3...$

► $0,1,\frac{1}{2},\frac{1}{4},...$

►

► None of these

Question No: 7 (Marks: 1) - Please choose one

If f is continuous on (a,b] but does not have a limit from the right then the integral

$$\int_a^b f(x)dx = \lim_{l\to a}\int_l^b f(x)dx$$

defined by is called ..... Integral

► Improper

► Proper

► None of these

Question No: 8 (Marks: 1) - Please choose one

An object is displaced 1m by a force of 1N then the work done W is

► 2

► 0

► None of these

► **1**

Question No: 9 (Marks: 1) - Please choose one

If f is a smooth function on [a,b] then the arc length L of the curve y=f(x) from x=a to x=b will be

► $L=\int_a^b \sqrt{1+[f'(x)]^2}\,dx$

$$L=\int_a^b \sqrt{1+[f'(x)]}dx$$

▶

$$L=\int_0^a \sqrt{1+[f'(x)]}dy$$

▶

▶ None of these

Question No: 10 (Marks: 1) - Please choose one

If f is a smooth function on [0,3] then the arc length L of the curve y=f(x) from x=0 to x=3 will be

$$L=\int_0^3 \sqrt{1+[f'(x)]^2}dx$$

▶

$$L=\int_a^b \sqrt{1+[f'(x)]^2}$$

▶

$$L=\int_0^3 \sqrt{1+[f'(x)]^2}dy$$

▶

▶ None of these

Question No: 11 (Marks: 1) - Please choose one

By using cylindrical shell to find volume of the solid when the region R in the first quadrant enclosed between $y=3x \text{ and } y=2x^2$ is revolved about the x-axis

$$V=\int_0^{\frac{3}{2}} 2\pi x(3x-2x^2)dx$$

▶

$$V = \int_0^{\frac{3}{2}} x(3x - 2x^2)dx$$

►

$$V = \int_0^{\frac{3}{2}} 2\pi(3x - 2x^2)dx$$

►

► None of these

Question No: 12 (Marks: 1) - Please choose one

By using cylindrical shell to find volume of the solid when the region R in the first quadrant enclosed between $y = x \text{ and } y = x^2$ is revolved about the y-axis is represented by

$$V = \int_0^3 2\pi x(x - x^2)dx$$

►

$$V = \int_0^1 x(x - x^2)dx$$

►

$$V = \int_0^1 2\pi(x - x^2)dx$$

►

► None of these

Question No: 13 (Marks: 1) - Please choose one

If

$$\int_a^a f(x)dx =$$

a is in the domain of f, then

► $f'(x)$

► $f(x)$

► **0**

► None of these

Question No: 14 (Marks: 1) - Please choose one

$\int_0^2 x^2 dx$

Consider the integral , the area on right is bounded by

► $y = x^2$

► $x = 2$

► $x = 0$

► None of these

Question No: 15 (Marks: 1) - Please choose one

The series $1-3+5-7+9-11$ may written as in sigma notation

► $\sum_{k=0}^{k=5}(-1)^k(2k+1)$

► $\sum_{k=1}^{k=5}(-1)^k(2k+1)$

► $\sum_{k=1}^{k=5}(2k+1)$

► None of these

Question No: 16 (Marks: 1) - Please choose one

$4^2+5^2+6^2+7^2$ in sigma notation may be represented as

$\sum_{k=2}^{k=7} k^2$

►

$\sum_{k=2}^{k=7} (k+1)^2$

►

$\sum_{k=4}^{k=7} k^2$

►

► None of these

Question No: 17 (Marks: 1) - Please choose one

If a function f is ..... on a closed interval [a,b] ,then f has both a maximum and minimum value on [a,b]

► **Continuous**
► Discontinuous
► Differentiable
► None of these

Question No: 18 (Marks: 1) - Please choose one

Let f be a function on an interval, and $x_1 \text{ and } x_2$ denote the points in that interval, if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ then the we can say that f is

► Increasing function
► **Decreasing function**
► Constant function
► None of these

Question No: 19 (Marks: 1) - Please choose one

If a function satisfies the conditions

f(c) is defined

$\lim_{x \to c^+} f(x)$

Exists

$\lim_{x \to c^+} f(x) = f(c)$

Then the function is said to be

► **Continuous at c**
► Continuous from left at c
► Continuous from right at c
► None of these

Question No: 20 (Marks: 1) - Please choose one

For a function f(x) to be continuous on interval (a,b) the function must be continuous

► **At all point in (a,b)**
► Only at a and b
► At mid point of a and b
► None of these

Question No: 21 (Marks: 2)

$$a_{n+1} = \frac{1}{3}(a_n + \frac{1}{a_n}) \; for \; n \geq 1 \; and \; a_1 = 2$$

Write down the first two term of the sequence

Question No: 22 (Marks: 2)

Find the integral of the surface area of the portion of the sphere generated by revolving the

$$y = \sqrt{2 - x^2} \; , 0 \leq x \leq \frac{1}{3}$$

curve
(Note: Just find the integral do not solve the integral)

Question No: 23 (Marks: 2)

$\int_2^5 f(x)dx$ $\int_2^3 f(x)dx = 7, \int_3^4 f(x)dx = 2, \int_5^4 f(x)dx = 5$

Calculate if

$\int_2^3 f(x)dx = 7, \int_3^4 f(x)dx = 2, \int_5^4 f(x)dx = 5$

$\int_2^5 f(x)dx = 7 + 2 - 5 = 4$

Question No: 24 (Marks: 3)

Find the first two Taylor polynomials for $\ln x$ about $x = 3$

Question No: 25 (Marks: 3)

Let the curve $y = x^{\frac{3}{2}} \;;\; 0 \le y \le 2$, then find the surface area generated by revolving the curve. (But do not evaluate)

Question No: 26 (Marks: 3)

$\frac{1}{1} + \frac{1}{4} + \frac{1}{9} + ... + \frac{1}{7225}$

Express the sum in sigma notation but do not evaluate.

$\frac{1}{1} + \frac{1}{4} + \frac{1}{9} + ... + \frac{1}{7225}$

$\sum_{k=1}^{7225} k^3 + 1$

Question No: 27 (Marks: 5)

Find the first four nonzero terms of the Taylor series generated by f at $x = a$

$f(x) = \frac{1}{1-x} \quad at \quad x = 2$

Question No: 28 (Marks: 5)

Evaluate the Definite Integral using the First fundamental Theorem of Calculus

$$\int_0^1 (x^5 - x^3 + 2x)dx$$

$$Let\ u = (x^5 - x^3 + 2x)$$

$$\int_0^1 (u)dx$$

Question No: 29 (Marks: 5)

Express the definite integrals as limits (Do not evaluate the integrals)

$$\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} (1 + Cosx)\, dx$$

$$\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} (1 + Cosx)\, dx$$

$$\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \lim_{\max xk \to 0} \sum_{\frac{-\pi}{2}}^{\frac{\pi}{2}} (1 + \cos x)dx$$

Question No: 30 (Marks: 10)

Find the region enclosed by the curves and also find the area

$$y = x^2\ ,\ y = \sqrt{x}\ ,\ x = \frac{1}{4}\ ,\quad x = 1$$

Question No: 31 (Marks: 10)

Use x^*_k as the left end point of each subinterval to find the area under $y = mx \;\; over\, the \,\text{int}\, erval\,[a,b], where\, m > 0\, and\, a \geq 0$

Solution on next page

Suppose $a = 1\ b = 2\ so [a,b] = [1,2]$

$x_k{}^* = x_{k-1} = a + (k-1)\Delta x \quad (formula\ for\ left\ end\ po\,\text{int})$

$$\Delta x = \frac{b-a}{n} = \frac{2-1}{n} = \frac{1}{n}$$

Suppose kth has area

$$f(x_{k^*})\Delta x = x_{k^*}\Delta x$$

$$\left[1+\frac{k}{n}\right]\Delta x$$

$$\left[1+\frac{k}{n}\right]\frac{1}{n}$$

$$\sum_{k=1}^{n} f(x_{k^*})\Delta x = \sum_{k=1}^{n}[1+k-1]$$

Area by solving

$$A = \lim_{\delta x \to 0}\sum_{k=1}^{n} f(x_{k^*})\Delta x = \lim_{\delta x \to 0}\left[\frac{3}{2}-1+\frac{1}{2n}\right]$$

$$= \frac{3}{2}-1+0$$

FINALTERM EXAMINATION 2009
(Session - 2)

Calculus & Analytical Geometry-I

Question No: 1 (Marks: 1) - Please choose one

According to Power-Rule of differentiation, if $f(x) = x^n$ where n is a real number, then

$\frac{d}{dx}[x^n] =$

- ▶ x^{n-1}
- ▶ $n\,x^{n-1}$
- ▶ $n\,x^{n+1}$
- ▶ $(n-1)x^{n+1}$

Question No: 2 (Marks: 1) - Please choose one

If a function g is differentiable at a point x and a function f is differentiable at a point g(x), then the _____ is differentiable at point x .

- ▶ Composition (f o g)
- ▶ Quotient (f / g)
- ▶ Product (f . g)
- ▶ Sum (f + g)

Question No: 3 (Marks: 1) - Please choose one

$y = f\left(g\left(h(x)\right)\right)$

If

$u = g\left(h(x)\right)$

$v = h(x)$ $\frac{dy}{dx} =$ _____

then

- ▶ $\frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$
- ▶ $\frac{dy}{du}\frac{du}{dv}\frac{dv}{dx}$
- ▶ $\frac{dv}{du} \cdot \frac{du}{dv} \cdot \frac{dy}{dx}$

Question No: 4 (Marks: 1) - Please choose one

If a function f is ..... on a closed interval [a,b] , then f has both maximum and minimum value on [a,b].

- ▶ Continuous
- ▶ Discontinuous
- ▶ None of these

Question No: 5 (Marks: 1) - Please choose one

$$\int_a^x \frac{t^2}{2} dt$$

The expression , represents a function of :

- ▶ t
- ▶ a
- ▶ Both x and a
- ▶ x

Question No: 6 (Marks: 1) - Please choose one

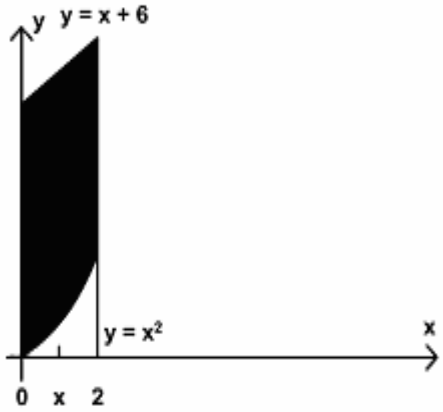
$\int cf(x)dx =$ _____

if c is a constant

- 0
- c
- $\int f(cx)dx$
- $c\int f(x)dx$

Question No: 7 (Marks: 1) - Please choose one

In the following figure, the area enclosed is bounded below by **:**



- $y = x + 6$
- $y = x^2$
- $x = 2$
- $x = 0$

Question No: 8 (Marks: 1) - Please choose one

At what points the two curves: $y = x^2 \ and \ y = x + 6$ intersect ?

- $x = 0 \ and \ x = 2$
- $x = 0 \ and \ x = 3$
- $x = 2 \ and \ x = 3$
- $x = -2 \ and \ x = 3$

Question No: 9 (Marks: 1) - Please choose one

What is the sum of following series?

$1+2+3+4+_____+n$

- $\frac{n+1}{2}$
- $\frac{(n+1)(n+2)}{2}$
- $\frac{n(n+2)}{2}$
- $\frac{n(n+1)}{2}$

Question No: 10 (Marks: 1) - Please choose one

$If\ b>0\ then\ \frac{d}{dx}[b^x] = _____$

- 0
- xb^{x-1}
- $\ln b$
- $b^x \ln b$

Question No: 11 (Marks: 1) - Please choose one

Let S be a solid bounded by two parallel planes perpendicular to the x-axis at x = a and x = b .If , for each x in [a,b], the cross-section area of S perpendicular to the x-axis is A(x) ,then what is the volume of the solid?

- $V=\int_a^b A(y)dx$
- $V=\int_a^b A(x)dx$

$$V = \int_{0}^{A(x)} [b-a]dx$$

▶

$$V = \int_{0}^{A(x)} [b+a]dx$$

▶

Question No: 12 (Marks: 1) - Please choose one

Let the solid generated by the region enclosed between $y = \sqrt{x}$; $x = 1, x = 4$ and the x-axis is revolved about the y-axis. Which of the following equation gives the volumes of a solid by cylindrical shells?

$$V = \int_{1}^{4} 2\pi x\sqrt{x}dx$$

▶

$$V = \int_{1}^{4} 2x\sqrt{x}dx$$

▶

$$V = \int_{0}^{4} 2x\sqrt{x}dx$$

▶

$$V = \int_{-4}^{4} 2x\sqrt{x}dx$$

▶

Question No: 13 (Marks: 1) - Please choose one

If slope m of a none vertical line is m =1 then the angle of inclination of the line is

$$\frac{\pi}{4}$$

▶

► $\frac{\pi}{2}$

► $\frac{\pi}{5}$

Question No: 14 (Marks: 1) - Please choose one

The PYTHAGORAS theorem describes the relationship between the sides of .....

► Right angle triangle
► Isoceleous triangle
► Equilateral triangle

Question No: 15 (Marks: 1) - Please choose one

If a quantity y depends on another quantity x in such a way that each value of x determines exactly one value of y , we say that y is ..... of x

► relation
► function
► not a function
► not a relation

Question No: 16 (Marks: 1) - Please choose one

The graph of the equation $y = x^2 - 4x + 5$ will represent

► Parabola
► Straight line
► Two straight lines

► Ellipse

Question No: 17 (Marks: 1) - Please choose one

The $\lim_{x \to a} f(x) = ...where\ f(x) = k$ (k is a constant)

► k+2
► k+1
► k

Question No: 18 (Marks: 1) - Please choose one

$$\int \frac{1}{x^2+2}\, dx$$

Consider the indefinite integral

Let $t = x^2 + 2$

Is the following substitution correct?

$\int \frac{1}{x^2+2}\, dx = \int \frac{1}{t}\, dt$

► Yes
► No

Question No: 19 (Marks: 1) - Please choose one

$\log_b \frac{1}{t} =$ _____

► $\log_b t$
► $1 - \log_b t$
► $1 + \log_b t$
► $-\log_b t$

Question No: 20 (Marks: 1) - Please choose one

How the series $1-3+5-7+9-11$ can be expressed in sigma notation?

► $\sum_{k=0}^{k=5} (-1)^k (2k+1)$

$$\sum_{k=1}^{k=5}(-1)^k(2k+1)$$

►

$$\sum_{k=1}^{k=5}(2k+1)$$

►

$$\sum_{k=1}^{k=5}(2k+1)$$

►

Question No: 21 (Marks: 1) - Please choose one

$$\sum_{k=1}^{n} f(x^*_k)\Delta x_k$$

The sum is known as:

► Riemann Sum
► General Sum
► Integral Sum
► Geometric Sum

Question No: 22 (Marks: 1) - Please choose one

$$\sum_{k=1}^{n} f(x^*_k)\Delta x_k$$

What does 'n' represent in the Riemann Sum ?

► No. of Circles
► No. of Subintervals
► No. of Loops
► No. of Squares

Question No: 23 (Marks: 1) - Please choose one

If *w* and *v* are continuous functions of *y* on an interval [c, d] and $w(y) \geq v(y) \ for \ c \leq y \leq d$, then area is bounded by the lines parallel to: **:**

► Y-axis
► X-axis
► Both X-axis and Y-axis

Question No: 24 (Marks: 1) - Please choose one

How the series $2(1)+2(2)+2(3)+2(4)+2(5)$ can be expressed in sigma notation?

- $\sum_{k=0}^{5} 2k^2$
- $\sum_{k=1}^{5} 2k^2$
- $\sum_{k=0}^{5} 2k$
- $\sum_{k=1}^{5} 2k$

Question No: 25 (Marks: 1) - Please choose one

$\sum_{k=1}^{n} \frac{k^3}{2} =$ _____

- $\frac{n(n+1)}{4}$
- $\frac{[n(n+1)]^2}{8}$
- $\frac{n(n+1)(2n+1)}{12}$
- $\frac{(n+1)(2n+1)}{6}$

Question No: 26 (Marks: 1) - Please choose one

If $a_1 < a_2 < a_3 < ..... < a_n < ....$, then a sequence $\{a_n\}$ is.....

- Nondecreasing
- Decreasing

► Increasing
► Nonincreasing

Question No: 27 (Marks: 1) - Please choose one

If $a_1 \geq a_2 \geq a_3 \geq ..... \geq a_n \geq ....$, then a sequence $\{a_n\}$ is .....

► Increasing
► Nondecreasing
► Decreasing
► Nonincreasing

Question No: 28 (Marks: 1) - Please choose one

If the difference between successive terms $a_{n+1} - a_n > 0$ then the sequence $\{a_n\}$ is known as**:**

► Increasing
► Decreasing
► Nondecreasing
► Nonincreasing

Question No: 29 (Marks: 1) - Please choose one

For a sequence $\{a_n\}$ if the ratio of successive terms $\frac{a_{n+1}}{a_n} > 1$ then the sequence is known as**:**

► Increasing
► Decreasing
► Nondecreasing
► Nonincreasing

Question No: 30 (Marks: 1) - Please choose one

For a sequence $\{a_n\}$ if the ratio of successive terms $\frac{a_{n+1}}{a_n} \geq 1$ then the sequence is known as **:**

► Increasing
► Decreasing
► Nondecreasing

► Nonincreasing

Question No: 31 (Marks: 1) - Please choose one

If $f(n) = a_n$ is the nth term of the sequence and $f(n)$ is differentiable and $f'(n) > 0$ then the sequence will be**:**

► Increasing
► Decreasing
► Nondecreasing
► Nonincreasing

Question No: 32 (Marks: 1) - Please choose one

If the geometric series $a + ar + ar^2 + ar^3 + ... + ar^{k-1} + ... \ \ where\ (a \neq 0)$, $|r| < 1$ then which of the following is true for the given series?

► Converges
► Diverges
► Gives no information

Question No: 33 (Marks: 1) - Please choose one

If the geometric series $a + ar + ar^2 + ar^3 + ... + ar^{k-1} + ... \ \ where\ (a \neq 0)$, $|r| \geq 1$ then which of the following is true for the given series?

► Converges
► Diverges
► Gives no information

Question No: 34 (Marks: 1) - Please choose one

If the series $\sum_{k=1}^{\infty} u_k$ converges but the series $\sum_{k=1}^{\infty} |u_k|$ does not converge , then $\sum_{k=1}^{\infty} u_k$ will _____

► Converge absolutely
► Diverge
► Converge conditionally

Question No: 35 (Marks: 1) - Please choose one

Let $\sum u_k$ be a series with nonzero terms and suppose that

$$\rho = \lim_{k\to\infty} \frac{|u_{k+1}|}{|u_k|} > 1$$

then which of the following is true?

► The series $\sum |u_k|$ converges

► The series $\sum |u_k|$ diverges

► No conclusion can be drawn.

Question No: 36 (Marks: 1) - Please choose one

Suppose f and g are integrable functions on [*a*, *b*] and *c* is a constant, then

$\int_a^b c\,[f(x)+g(x)]\,dx =$ _____

► $\int_a^b f(cx)dx + \int_a^b g(cx)dx$

► $\int_a^b f(x)\,dx + \int_a^b g(x)]\,dx$

► $c\int_a^b f(x)dx + c\int_a^b g(x)dx$

►

► 0

Question No: 37 (Marks: 1) - Please choose one

Which of the following is surface area S generated by revolving the curve *y = f(x)* between *x = 0* and *x = 2* about the *x*-axis?

$$S = \int_0^2 2\pi f(x)\sqrt{1+[f(x)]}dx$$

▶

$$S = \int_0^2 2\pi f(x)\sqrt{1+[f'(x)]^2}dx$$

▶

$$S = \int_0^2 2\pi f(x)\sqrt{1+[f'(x)]}dx$$

▶

$$S = \int_0^2 2\sqrt{1+[f'(x)]}dx$$

▶

Question No: 38 (Marks: 1) - Please choose one

Which of the following is area of the surface generated by revolving the curve $y = 4\sqrt{x}\ ;\ 1 \le x \le 4$ about the *x*-axis?

$$\int_1^4 2\pi\left(4\sqrt{x}\right)\sqrt{1+\left[\left(4\sqrt{x}\right)\right]^2}\,dx$$

▶

$$\int_1^4 2\pi\left(4\sqrt{x}\right)\sqrt{1+\left[\left(4\sqrt{x}\right)'\right]}\,dx$$

▶

$$\int_1^4 2\pi+\sqrt{1+\left[\left(4\sqrt{x}\right)'\right]^2}\,dx$$

▶

$$\int_1^4 2\pi\left(4\sqrt{x}\right)\sqrt{1+\left[\left(4\sqrt{x}\right)'\right]^2}\,dx$$

▶

Question No: 39 (Marks: 1) - Please choose one

Which of the following is the work done *W* if an object moves in the positive direction along a coordinate line while subject to a force *F(x)* in the direction of motion over an interval [0,3]?

$$W = \int_{2}^{3} 3xdx$$

▶

$$W = \int_{0}^{3} 3xdx$$

▶

$$W = \int_{0}^{3} F(x)dx$$

▶

$$W = \int_{3}^{0} F(x)dx$$

▶

Question No: 40 (Marks: 1) - Please choose one

$\int_{1}^{0} f(x)\,dx = 2$ $\int_{0}^{5} f(x)\,dx = 1$

If and then which of the following is value of $\int_{1}^{5} f(x)\,dx$?

▶ -3
▶ -1
▶ 1
▶ 3

Question No: 41 (Marks: 2)

Derivative of a function is $\frac{1}{2\sqrt{x}}$. Find the original function.

Question No: 42 (Marks: 2)

If $\sum u_k$ is a series with positive terms and $\rho = \lim_{k \to +\infty} \sqrt[k]{u_k}$; then write the three cases for the series to be convergent, divergent or none.

Question No: 43 (Marks: 2)

Evaluate the following integral:

$$\int_1^3 \frac{1}{x^2}\,dx$$

Question No: 44 (Marks: 3)

Use the first fundamental theorem of calculus to evaluate the definite integral:

$$\int_0^2 f(x)\,dx \qquad where \qquad f(x)=\begin{cases} x^2 & ; \quad 0 \le x < 1 \\ x^3 & ; \quad 1 \le x \le 2 \end{cases}$$

Question No: 45 (Marks: 3)

$$\sum_{k=2}^{\infty}(-1)^{k-1}\frac{2^{k-1}}{(k-1)!}$$

Show that the series converges absolutely.

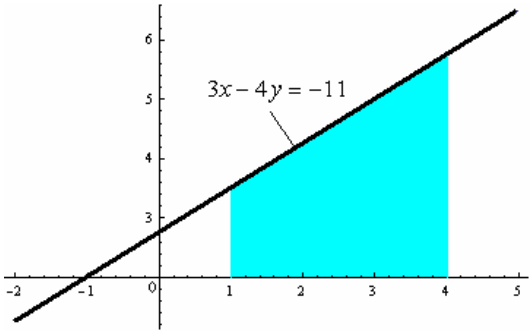
Question No: 46 (Marks: 3)

Express the following definite integral as limit of Riemann Sum. (Do not evaluate the integral)

$$\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} Cos\,x\,dx$$

Question No: 47 (Marks: 5)

Express area of the shaded region as a definite integral.



Question No: 48 (Marks: 5)

How much work is required to wind the chain onto the pulley if a 100-ft length of steel chain weighting
15 lb/ ft. is dangling from a pulley?

Question No: 49 (Marks: 5)

Evaluate the following integral:

$$\int_1^2 \frac{x^2+\sqrt{x}}{x^2}\,dx$$

Question No: 50 (Marks: 10)

Use L'Hopital's Rule to evaluate

$$\lim_{x\to 0}(1+\sin x)^{\cot x}$$

FINALTERM EXAMINATION 2009
(Session - 1)

Calculus & Analytical Geometry-I

Question No: 1 (Marks: 1) - Please choose one

If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) > 0$ then f has relative ..... At x_0

► Minima
► Maxima
► None of these

Question No: 2 (Marks: 1) - Please choose one

If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) < 0$ then f has relative ..... At x_0

► Minima
► Maxima
► None of these

Question No: 3 (Marks: 1) - Please choose one

$\lim_{x \to 0} \frac{\sin 2x}{x} = - - - - -$

► 2
► 4

- 1
- ∞

Question No: 4 (Marks: 1) - Please choose one

$$\lim_{x\to 0^+} \frac{\ln x}{1/x} = -----$$

- 1
- 0
- e
- None of these

Question No: 5 (Marks: 1) - Please choose one

$$\frac{d(\tan x)}{dx} =$$

- $\sec x$
- $\sec^2 x$
- $co\sec x$
- $co\sec^2 x$

Question No: 6 (Marks: 1) - Please choose one

If $xy = 4$ then $\frac{dy}{dx} =$

- 0
- $\frac{-1}{x^2}$
- $\frac{4}{x^2}$
- $\frac{-4}{x^2}$

Question No: 7 (Marks: 1) - Please choose one

Consider a function $h(x)$ and a constant c then

$\frac{d}{dx}\left((c)\{h(x)\}\right) =$ _____

- 0
- $\frac{d}{dx}\left(h(x)\right)$
- $\frac{d}{dx}\left(h(cx)\right)$
- $c\frac{d}{dx}\left(h(x)\right)$

Question No: 8 (Marks: 1) - Please choose one

Suppose that f and g are differentiable functions of x then

$\frac{d}{dx}[f][g] =$

- $\frac{[f'][g]-[f][g']}{g^2}$
- $[f'][g']$
- $[f'][g]+[f][g']$
- $[f'][g]-[f][g']$

Question No: 9 (Marks: 1) - Please choose one

The power rule, $\frac{d}{dx}[x^n] = nx^{n-1}$ holds if n is _____

- An integer
- A rational number
- An irrational number
- All of the above

Question No: 10 (Marks: 1) - Please choose one

Let a function f be defined on an interval, and let x_1 and x_2 denotes two distinct points in that interval. If $f(x_1) = f(x_2)$ for all points x_1 and x_2 then

which of the following statement is correct?

- ▶ f is a decreasing function
- ▶ f is an increasing function
- ▶ f is a constant function

Question No: 11 (Marks: 1) - Please choose one

If $f''(x) < 0$ on an open interval (a,b) then which of the following statement is correct?

- ▶ f is concave up on (a, b).
- ▶ f is concave down on (a, b)
- ▶ f is linear on (a, b).

Question No: 12 (Marks: 1) - Please choose one

$$\sum_{k=1}^{n} f(x^*_k)\Delta x_k$$

What does 'n' represent in Riemann Sum ?

- ▶ No. of Circles
- ▶ No. of Rectangles
- ▶ No. of Loops
- ▶ No. of Squares

Question No: 13 (Marks: 1) - Please choose one

$$\lim_{x\to-\infty} f(x) = +\infty \quad and \quad \lim_{x\to+\infty} f(x) = +\infty$$

If f is continuous function such that $(-\infty, +\infty)$ then f has _____ on

- ▶ maximum value but no minimum
- ▶ minimum value but no maximum
- ▶ both maximum and minimum value

Question No: 14 (Marks: 1) - Please choose one

$$\int_2^t \frac{x^2}{2} dx$$

The expression , represents a function of :

- ▶ t
- ▶ x
- ▶ 2
- ▶ Both t and x

Question No: 15 (Marks: 1) - Please choose one

$\int cf(x)dx =$ _____

if c is a constant

- 0
- c
- $\int f(cx)dx$
- $c\int f(x)dx$

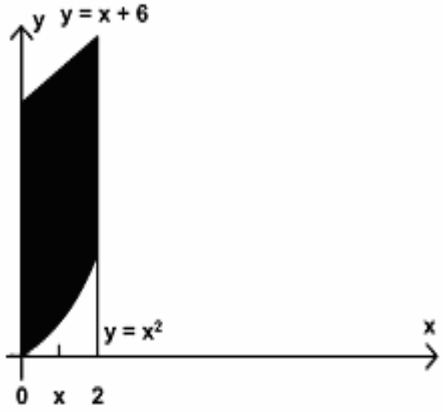
Question No: 16 (Marks: 1) - Please choose one

Sigma notation is represented by which of the following Greek letter?

- χ
- η
- Σ
- ψ

Question No: 17 (Marks: 1) - Please choose one

In the following figure, the area enclosed is bounded below by **:**

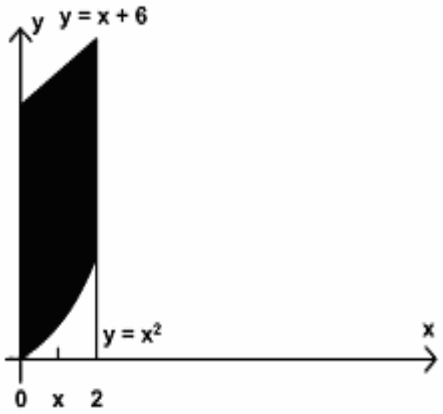


- $y = x + 6$
- $y = x^2$
- $x = 2$

- $x = 0$

Question No: 18 (Marks: 1) - Please choose one

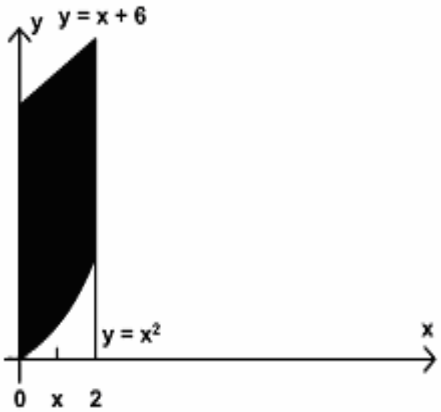
In the following figure, the area bounded on the sides by the lines are **:**



- $x = 0$
- $x = 2$
- $x = 0 \; and \; x = 2$
- $x = 6$

Question No: 19 (Marks: 1) - Please choose one

What is the area of the region in the following figure?



- $A = \int_{0}^{2} \left[(x+6) - (x^2) \right] dx$
- $A = \int_{x}^{2} \left[(x+6) - (x^2) \right] dx$

$$A=\int_0^2\left[(x+6)+\left(x^2\right)\right]dx$$

▶

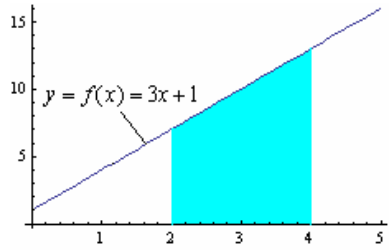
$$A=\int_0^x\left[(x+6)-\left(x^2\right)\right]dx$$

▶

Question No: 20 (Marks: 1) - Please choose one

Which of the following is approximate area under the curve $y=f(x)=3x+1$ over the interval $[2, 4]$, evaluated by using the formula

$$Area=f(x_1^*)\,\Delta x+f(x_2^*)\,\Delta x$$



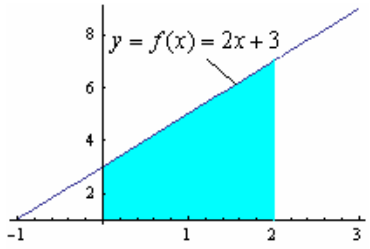
If the interval $[2, 4]$ is divided into two sub-intervals of equal length and x_1^* and x_2^* are left endpoint of each sub-interval.

▶ 17
▶ 20
▶ 23

Question No: 21 (Marks: 1) - Please choose one

Which of the following is approximate area under the curve $y=f(x)=2x+3$ over the interval $[0, 2]$, evaluated by using the formula

$$Area=f(x_1^*)\,\Delta x+f(x_2^*)\,\Delta x$$



If the interval $[0, 2]$ is divided into two sub-intervals of equal length and x_1^* and x_2^* are right endpoint of each sub-interval.

▶ 8
▶ 10
▶ 12

Question No: 22 (Marks: 1) - Please choose one

$If\ \ x>0\ then\ \frac{d}{dx}[\ln x]\ =$ _____

► 1

► x

► $\frac{1}{x}$

► $\ln \frac{1}{x}$

Question No: 23 (Marks: 1) - Please choose one

Suppose f and g are integrable functions on [a,b] and c is a constant, then $\int_a^b c\,[f(x)+g(x)]\,dx =$ _____

► $\int_a^b f(cx)dx + \int_a^b g(cx)dx$

► $\int_a^b f(x)\,dx + \int_a^b g(x)]\,dx$

► $c\int_a^b f(x)dx + c\int_a^b g(x)dx$

► 0

Question No: 24 (Marks: 1) - Please choose one

If the function f is continuous on [a,b] and if $f(x) \geq 0$ for all x in [a,b], then which of the following gives area under the curve $y = f(x)$ over the interval [a,b]?

► $\lim_{x\to\infty} \sum_{k=1}^{n} [x_k][f(x_k)]$ *where n is number of subdivisions of* $[a,b]$

► $\int_a^b f(x)\,dx$

► $\pi[radius]^2$

► (Width) (Height)

Question No: 25 (Marks: 1) - Please choose one

Let region R in the first quadrant enclosed between $y = 3x \quad and \quad y = 2x^2$ is revolved about the
x-axis .Which of the following equation gives the volume of a solid by cylindrical shells?

$V = \int_0^{\frac{3}{2}} 2\pi x(3x - 2x^2)dx$

▶

$V = \int_0^{\frac{3}{2}} x(3x - 2x^2)dx$

▶

$V = \int_0^{\frac{3}{2}} 2\pi(3x - 2x^2)dx$

▶

$V = \int_{-1}^{\frac{3}{2}} 2\pi(3x - 2x^2)dx$

▶

Question No: 26 (Marks: 1) - Please choose one

Let f is a smooth function on [a, b]. What will be the arc length L of the curve y = f(x) from x = a to x = b?

$L = \int_b^a \sqrt{1 + [f'(x)]}dy$

▶

$L = \int_a^b \sqrt{1 + [f'(x)]^2}dx$

▶

$$L=\int_0^a \sqrt{1+[f'(x)]}dy$$

►

$$L=\int_a^b \sqrt{1+[f'(x)]}dx$$

►

Question No: 27 (Marks: 1) - Please choose one

If f is continuous on (a, b] but does not have a limit from the right then the

$$\int_a^b f(x)dx=\lim_{l\to a}\int_l^b f(x)dx$$

integral defined by is called **:**

► Improper
► Proper
► Line

Question No: 28 (Marks: 1) - Please choose one

For a sequence $\{a_n\}$ if the ratio of successive terms $\frac{a_{n+1}}{a_n}>1$ then the sequence is known as**:**

► Increasing
► Decreasing
► Nondecreasing
► Nonincreasing

Question No: 29 (Marks: 1) - Please choose one

For a sequence $\{a_n\}$ if the ratio of successive terms $\frac{a_{n+1}}{a_n}<1$ then the sequence is known as**:**

► Increasing
► Decreasing
► Nondecreasing
► Nonincreasing

Question No: 30 (Marks: 1) - Please choose one

$$\int \frac{3x^2+4x+1}{x^3+2x^2+x-3}\,dx$$

Consider the indefinite integral

Let $t = x^3+2x^2+x-3$

Is the following substitution correct?

$$\int \frac{3x^2+4x+1}{x^3+2x^2+x-3}\,dx = \int \frac{1}{t}\,dt$$

- ► Yes
- ► No

Question No: 31 (Marks: 1) - Please choose one

$$\rho = \lim_{k\to\infty} \frac{u_{k+1}}{u_k}$$

The series $\sum u_k$ be a series with positive terms and suppose that

if $\rho = 1$, then which of the following is true?

- ► Converges
- ► Diverges
- ► May converges or diverges
- ► Gives no information

Question No: 32 (Marks: 1) - Please choose one

The series $\sum u_k$ be a series with positive terms and suppose that

$\rho = \lim_{k\to\infty} \sqrt[k]{u_k} = \lim_{k\to\infty} (u_k)^{\frac{1}{k}}$ if $\rho = 1$, then which of the following is true?

- ► Converges
- ► Diverges
- ► May converges or diverges
- ► Gives no information

Question No: 33 (Marks: 1) - Please choose one

$$\sum_{k=1}^{\infty} |u_k| = |u_1| + |u_2| + |u_3| + \ldots |u_k| + \ldots$$

If the series converges , then which of

$$\sum_{k=1}^{\infty} u_k = u_1 + u_2 + u_2 + \ldots u_k + \ldots$$

the following is true for ?

► Converges
► Diverges
► Gives no information

Question No: 34 (Marks: 1) - Please choose one

$$\rho = \lim_{k\to\infty} \frac{|u_{k+1}|}{|u_k|}$$

Let $\sum u_k$ be a series with nonzero terms and suppose that if $\rho = +\infty$, then which of the following is true?

► Then the series $\sum u_k$ diverges
► The series $\sum u_k$ converges absolutely and therefore converges
► May converges or diverges
► Gives no information

Question No: 35 (Marks: 1) - Please choose one

$$\int_{-1}^{1} (x-1)\, dx = _____$$

► -2
► 0
► 2
► 4

Question No: 36 (Marks: 1) - Please choose one

How many critical points exist for a function f if

$f'(x) = (x-3)(x-2)$

- Zero
- One
- Two
- Four

Question No: 37 (Marks: 1) - Please choose one

$\log_b ac =$ _____

- $\log_b a + \log_b c$
- $\log_b a - \log_b c$
- $\frac{\log_b a}{\log_b c}$
- $(\log_b a)(\log_b c)$

Question No: 38 (Marks: 1) - Please choose one

$\log_b a^r =$ _____

- $a \log_b r$
- $r \log_b a$
- $\frac{\log_b a}{\log_b r}$
- $\log_b a + \log_b r$

Question No: 39 (Marks: 1) - Please choose one

Let $y = \frac{2\sqrt{2}}{3} x^{\frac{3}{2}}$; $0 \le x \le 2$ then which of the following is the length of the curve?

$$L=\int_0^2\sqrt{\left[\frac{d}{dx}\left(\frac{2\sqrt{2}}{3}x^{\frac{3}{2}}\right)\right]^2}\,dx$$

▶

$$L=\int\sqrt{1+\left[\frac{d}{dx}\left(\frac{2\sqrt{2}}{3}x^{\frac{3}{2}}\right)\right]^2}\,dx$$

▶

$$L=\int_0^2\sqrt{1+\left[\frac{d}{dx}\left(\frac{2\sqrt{2}}{3}x^{\frac{3}{2}}\right)\right]^2}\,dx$$

▶

$$L=\int_0^2\sqrt{1+\left[\frac{d}{dx}\left(\frac{2\sqrt{2}}{3}x^{\frac{3}{2}}\right)\right]}\,dx$$

▶

Question No: 40 (Marks: 1) - Please choose one

Which of the following are *first two* terms for the Taylor series of $f(x)=e^{-x}$ at $x = 0$?

$1+(1)(x-0)$

▶

$1+(-1)(x+0)$

▶

$1+(-1)(x-0)$

▶

$(-1)(x-0)$

▶

Question No: 41 (Marks: 2)

$$\int_2^3(1-x)dx$$

Evaluate the integral

$$\int_2^3 (1-x)dx$$

$$= \left| x - x^2 \frac{1}{2} \right|_2^3$$

$$= \frac{1}{2} \left| 2x - x^2 \right|_2^3$$

$$= \frac{1}{2}(2(3-2)-(3-2)^2)$$

$$= \frac{1}{2}(2-1)$$

$$= \frac{1}{2}$$

Question No: 42 (Marks: 2)

$$\int_2^{+\infty} \frac{dx}{x^2}$$

Evaluate the improper integral

Question No: 43 (Marks: 2)

A function $f(x) = x^2 - 4x - 9$ has critical point 2 in an interval [0, 5]. Find the maximum value of the function and point having this value.

Question No: 44 (Marks: 3)

$$\int \frac{5 - 6\sin^2 x}{\sin^2 x} dx$$

Evaluate:

$$\int \frac{5 - 6\sin^2 x}{\sin^2 x} dx$$

Question No: 45 (Marks: 3)

Find the area of the region bounded by the curve $y = x^2 \quad , \quad x > 0$, and bounded on the sides by the lines $y = 1$ and $y = 4$

$y = x^2 \quad , \quad x > 0$

So we have

$$A = \int_1^4 x^2 dx$$
$$= \left| \frac{x^3}{3} \right|_1^4$$
$$= \frac{1}{3}(4-1)^3$$
$$= \frac{1}{3}(3)^3$$
$$= 9$$

Question No: 46 (Marks: 3)

Determine whether the following sequence converges or diverges. If it converges, find the limit.

$$\lim_{n\to\infty} \frac{5n^2 - 1}{20n + 7n^2}$$

Question No: 47 (Marks: 5)

Use the Alternating series Test to determine whether the given series converges

$$\sum_1^\infty \frac{(-1)^{n-1} \cdot n!}{2^n}$$

Question No: 48 (Marks: 5)

Evaluate the integral

$$\int_{\frac{\pi}{2}}^0 \frac{1+\cos 2t}{2} dt$$

Solution

$$\int_{\frac{\pi}{2}}^{0} \frac{1+\cos 2t}{2} dt$$

$$u = 2t$$

$$\frac{du}{dt} = 2dt$$

$$du = 2dt$$

$$so$$

$$= \frac{1}{4}\int_{\frac{\pi}{2}}^{0} 1+\cos u du$$

$$= \frac{1}{4} | u + \sin u |_{\frac{\pi}{2}}^{0}$$

$$= \frac{1}{4} | 2t + \sin 2t |_{\frac{\pi}{2}}^{0}$$

$$= \frac{1}{4}(2\frac{\pi}{2} + \sin 2\frac{\pi}{2})$$

$$= \frac{1}{4}(\pi + \sin \pi)$$

$$= \frac{1}{4}(\pi + 0)$$

$$= \frac{\pi}{4}$$

Question No: 49 (Marks: 5)

Evaluate the sums

$$\sum_{k=1}^{5} k(3k+5)$$

$$= 1(3+5) + 2(6+5) + 3(9+5) + 4(12+5) + 5(15+5)$$

$$= 8 + 22 + 3(45) + 4(60) + 5(75)$$

$$= 8 + 22 + 135 + 240 + 375$$

$$= 780$$

Question No: 50 (Marks: 10)

Find the volume of the solid that results when the region enclosed by the given curves is revolved about the x – axis**.**

$y = 1 + x^3$, $x = 1, x = 2, y = 0$

from $V = \int_a^b \pi[f(x)]^2 dx$

$$V = \int_1^2 \pi[1 + x^3]^2 dx$$

$$V = \int_1^2 \pi[1 + x^5 + 2x^3] dx$$

$$V = \pi \int_1^2 (1 + x^5 + 2x^3) dx$$

$$V = \pi \mid (x + \frac{1}{6}x^6 + \frac{1}{2}x^4) \mid_1^2$$

$$V = \pi((2-1) + \frac{1}{6}(2-1)^6 + \frac{1}{2}(2-1)^4)$$

$$V = \pi\{((2-1) + \frac{1}{6}(2-1)^6 + \frac{1}{2}(2-1)^4)\}$$

$$V = \pi((1) + \frac{1}{6} + \frac{1}{2}))$$

$$V = \frac{\pi(6+1+3)}{6}$$

$$V = \frac{\pi(10)}{6} = \pi\frac{5}{3}$$

This paper is solved by our best knowledge. In the case of any error/correction/suggestion, please contact at gulshanvu@yahoo.com, with reference to the concerned paper's number.

FINALTERM EXAMINATION
Fall 2008
(Session - 1)

Calculus & Analytical Geometry-I

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UPDATED VERSION
Exclusive thanks to Mahar Azahar (Lodhran)

Question No: 1 (Marks: 1) - Please choose one

If $y = f(x)$ then the average rate of change of y with respect to x over the interval $[x_0, x_1]$ is the ..... Joining the points $(x_0, f(x_0)) and (x_1, f(x_1))$ on the graph of f

- ► Slope of the secant line
- ► Slope of tangent line
- ► Secant line
- ► none of these

Question No: 2 (Marks: 1) - Please choose one

Natural domain of $\frac{(x^2-4)}{(x-2)}$ is

- ► $(-\infty, 2) U (2, +\infty)$
- ► $(-\infty, 2)$
- ► $(-\infty, 0)$
- ► None of these

vuZs

Question No: 3 (Marks: 1) - Please choose one

The equation $(x+4)^2+(y-1)^2=6$ represents a circle having center at ..... and radius .....

► $(-4,1),\sqrt{6}$

► $(-4,1),6$

► $(-4,-1),\sqrt{6}$

► None of these

Question No: 4 (Marks: 1) - Please choose one

The series $\sum u_k$ be a series with positive terms and suppose that $\rho=\lim_{k\to\infty}\sqrt[k]{u_k}=\lim_{k\to\infty}(u_k)^{\frac{1}{k}}$ if $\rho>1$ then the series

VUZs

► Converges

► **Diverges**

► May converge or diverge

► None of these

Question No: 5 (Marks: 1) - Please choose one

The series $\sum u_k \text{ and } \sum v_k$ are convergent series then $(\sum u_k+\sum v_k) \text{ and } (\sum u_k-\sum v_k)$ will be .....and.....

► Convergent, convergent

► Divergent, divergent

► **Convergent, divergent**

► Divergent, convergent

Question No: 6 (Marks: 1) - Please choose one

The notation $\{\frac{1}{2^n}\}_{-1}^{n}$ represents the sequence

► $2,1,\frac{1}{2},\frac{1}{4},...$

► $0,1,2,3...$

► $0,1,\frac{1}{2},\frac{1}{4},...$

►

► None of these

Question No: 7 (Marks: 1) - Please choose one

If f is continuous on (a,b] but does not have a limit from the right then the integral

$$\int_a^b f(x)dx = \lim_{l\to a}\int_l^b f(x)dx$$

defined by is called ..... Integral

► Improper

► Proper

► None of these

Question No: 8 (Marks: 1) - Please choose one

An object is displaced 1m by a force of 1N then the work done W is

► 2

► 0

► None of these

► **1**

Question No: 9 (Marks: 1) - Please choose one

If f is a smooth function on [a,b] then the arc length L of the curve y=f(x) from x=a to x=b will be

► $L=\int_a^b \sqrt{1+[f'(x)]^2}\,dx$

vuZs

$$L=\int_{a}^{b}\sqrt{1+[f'(x)]}dx$$

▶

$$L=\int_{0}^{a}\sqrt{1+[f'(x)]}dy$$

▶

▶ None of these

Question No: 10 (Marks: 1) - Please choose one

If f is a smooth function on [0,3] then the arc length L of the curve y=f(x) from x=0 to x=3 will be

$$L=\int_{0}^{3}\sqrt{1+[f'(x)]^2}dx$$

▶

$$L=\int_{a}^{b}\sqrt{1+[f'(x)]^2}$$

▶

$$L=\int_{0}^{3}\sqrt{1+[f'(x)]^2}dy$$

▶

▶ None of these

Question No: 11 (Marks: 1) - Please choose one

By using cylindrical shell to find volume of the solid when the region R in the first quadrant enclosed between $y=3x \text{ and } y=2x^2$ is revolved about the x-axis

$$V=\int_{0}^{\frac{3}{2}}2\pi x(3x-2x^2)dx$$

▶

$$V = \int_0^{\frac{3}{2}} x(3x - 2x^2)dx$$

►

$$V = \int_0^{\frac{3}{2}} 2\pi(3x - 2x^2)dx$$

►

► None of these

Question No: 12 (Marks: 1) - Please choose one

By using cylindrical shell to find volume of the solid when the region R in the first quadrant enclosed between $y = x \text{ and } y = x^2$ is revolved about the y-axis is represented by

$$V = \int_0^3 2\pi x(x - x^2)dx$$

►

$$V = \int_0^1 x(x - x^2)dx$$

►

$$V = \int_0^1 2\pi(x - x^2)dx$$

►

► None of these

Question No: 13 (Marks: 1) - Please choose one

If

$$\int_a^a f(x)dx =$$

a is in the domain of f, then

► $f'(x)$

► $f(x)$

► **0**

► None of these

Question No: 14 (Marks: 1) - Please choose one

$$\int_0^2 x^2 dx$$

Consider the integral , the area on right is bounded by

► $y = x^2$

► $x = 2$

► $x = 0$

► None of these

Question No: 15 (Marks: 1) - Please choose one

The series $1-3+5-7+9-11$ may written as in sigma notation

► $\sum_{k=0}^{k=5}(-1)^k(2k+1)$

► $\sum_{k=1}^{k=5}(-1)^k(2k+1)$

► $\sum_{k=1}^{k=5}(2k+1)$

► None of these

vuZs

Question No: 16 (Marks: 1) - Please choose one

$4^2+5^2+6^2+7^2$ in sigma notation may be represented as

► $\sum_{k=2}^{k=7} k^2$

► $\sum_{k=2}^{k=7} (k+1)^2$

► $\sum_{k=4}^{k=7} k^2$

►

► None of these

Question No: 17 (Marks: 1) - Please choose one

If a function f is ..... on a closed interval [a,b] ,then f has both a maximum and minimum value on [a,b]

► **Continuous**

► Discontinuous

► Differentiable

► None of these

Question No: 18 (Marks: 1) - Please choose one

Let f be a function on an interval, and $x_1 \text{ and } x_2$ denote the points in that interval, if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ then the we can say that f is

vuZs

► Increasing function

► **Decreasing function**

► Constant function

► None of these

Question No: 19 (Marks: 1) - Please choose one

If a function satisfies the conditions

f(c) is defined

$\lim_{x \to c^+} f(x)$

Exists

$\lim_{x \to c^+} f(x) = f(c)$

Then the function is said to be

► **Continuous at c**
► Continuous from left at c
► Continuous from right at c
► None of these

Question No: 20 (Marks: 1) - Please choose one

For a function f(x) to be continuous on interval (a,b) the function must be continuous

► **At all point in (a,b)**
► Only at a and b
► At mid point of a and b
► None of these



Question No: 21 (Marks: 2)

$$a_{n+1} = \frac{1}{3}(a_n + \frac{1}{a_n}) \; for \; n \geq 1 \; and \; a_1 = 2$$

Write down the first two term of the sequence

Question No: 22 (Marks: 2)

Find the integral of the surface area of the portion of the sphere generated by revolving the

$$y = \sqrt{2 - x^2} \; , 0 \leq x \leq \frac{1}{3}$$

curve
(Note: Just find the integral do not solve the integral)

Question No: 23 (Marks: 2)

$\int_2^5 f(x)dx$ $\int_2^3 f(x)dx = 7, \int_3^4 f(x)dx = 2, \int_5^4 f(x)dx = 5$

Calculate if

$\int_2^3 f(x)dx = 7, \int_3^4 f(x)dx = 2, \int_5^4 f(x)dx = 5$

$\int_2^5 f(x)dx = 7 + 2 - 5 = 4$

Question No: 24 (Marks: 3)

Find the first two Taylor polynomials for $\ln x$ about $x = 3$

Question No: 25 (Marks: 3)

Let the curve $y = x^{\frac{3}{2}} \;;\; 0 \le y \le 2$, then find the surface area generated by revolving the curve. (But do not evaluate)

Question No: 26 (Marks: 3)

$\frac{1}{1} + \frac{1}{4} + \frac{1}{9} + ... + \frac{1}{7225}$

Express the sum in sigma notation but do not evaluate.

$\frac{1}{1} + \frac{1}{4} + \frac{1}{9} + ... + \frac{1}{7225}$

$\sum_{k=1}^{7225} k^3 + 1$

Question No: 27 (Marks: 5)

Find the first four nonzero terms of the Taylor series generated by f at $x = a$

$f(x) = \frac{1}{1-x} \quad at \quad x = 2$

Question No: 28 (Marks: 5)

Evaluate the Definite Integral using the First fundamental Theorem of Calculus

$$\int_0^1 (x^5 - x^3 + 2x)dx$$

$$Let\ u = (x^5 - x^3 + 2x)$$

$$\int_0^1 (u)dx$$

vuZs

Question No: 29 (Marks: 5)

Express the definite integrals as limits (Do not evaluate the integrals)

$$\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} (1+Cosx)\,dx$$

$$\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} (1+Cosx)\,dx$$

$$\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \lim_{\max xk \to 0} \sum_{\frac{-\pi}{2}}^{\frac{\pi}{2}} (1+\cos x)dx$$

vuZs

Question No: 30 (Marks: 10)

Find the region enclosed by the curves and also find the area

$$y = x^2\ ,\ y = \sqrt{x}\ ,\ x = \frac{1}{4}\ ,\quad x = 1$$

Question No: 31 (Marks: 10)

Use x^*_k as the left end point of each subinterval to find the area under $y = mx \ \ over\ the \text{int} erval [a,b], where\ m > 0\ and\ a \geq 0$

Solution on next page

Suppose $a = 1\ b = 2\ so [a,b] = [1,2]$

$x_k * = x_{k-1} = a + (k-1)\Delta x \quad (formula\ for\ left\ end\ po\text{int})$

$$\Delta x = \frac{b-a}{n} = \frac{2-1}{n} = \frac{1}{n}$$

Suppose kth has area

$$f(x_{k*})\Delta x = x_{k*}\Delta x$$

$$\left[1 + \frac{k}{n}\right]\Delta x$$

$$\left[1 + \frac{k}{n}\right]\frac{1}{n}$$

$$\sum_{k=1}^{n} f(x_{k*})\Delta x = \sum_{k=1}^{n}[1 + k - 1]$$

Area by solving

$$A = \lim_{\delta x \to 0} \sum_{k=1}^{n} f(x_{k*})\Delta x = \lim_{\delta x \to 0}\left[\frac{3}{2} - 1 + \frac{1}{2n}\right]$$

$$= \frac{3}{2} - 1 + 0$$

This paper is solved by our best knowledge. In the case of any error/correction/suggestion, please contact at gulshanvu@yahoo.com, with reference to the concerned paper's number.

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FINALTERM EXAMINATION
(Session - 4)

Calculus & Analytical Geometry-I

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UPDATED VERSION
Exclusive thanks to Mahar Azahar (Lodhran)

(Marks: 1) - Please choose one

Let the region bounded by the curve $y = \sqrt[3]{x}$, the x-axis, and the line $x = 3$ is revolved about the y-axis to generate a solid. Which of the following equation gives the volume of a solid by cylindrical shells?

$$V = \int_0^3 x^{\frac{3}{2}}\, dx$$

▶

$$V = 2\pi \int_0^3 \sqrt{x}\, dx$$

▶

$$V = \int_0^3 2\pi x \sqrt[3]{x}\, dx$$

▶

$$V = \int_0^3 x \sqrt[3]{x}\, dx$$

▶

(Marks: 1) - Please choose one

For a sequence $\{a_n\}$ if the difference between successive terms $a_{n+1} - a_n \leq 0$ then the sequence is known as**:**

► Increasing
► Decreasing
► Nondecreasing
► Nonincreasing

(Marks: 1) - Please choose one

For a sequence $\{a_n\}$ if the ratio of successive terms $\frac{a_{n+1}}{a_n} < 1$ then the sequence is known as**:**

► Increasing
► Decreasing
► Nondecreasing
► Nonincreasing

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(Marks: 1) - Please choose one

If $\rho = \lim_{k \to +\infty} \sqrt[k]{u_k}$ where $\rho > 1$ then the series $\sum u_k$ with positive terms will /will be.....?

► Convergent
► Divergent
► Give no information

(Marks: 1) - Please choose one

In alternating series test, which one of the following condition must be satisfied?

► $\lim_{k \to \infty} a_k = 1$
► $a_1 > a_2 > a_3 ..... > a_k > ....$
► $a_1 \le a_2 \le a_3 ..... \le a_k \le ....$

(Marks: 1) - Please choose one

Let $y = \frac{2\sqrt{2}}{3} x^{\frac{3}{2}} \; ; \; 0 \le x \le 2$ then which of the following is the length of the curve?

$$L=\int_0^2\sqrt{\left[\frac{d}{dx}\left(\frac{2\sqrt{2}}{3}x^{\frac{3}{2}}\right)\right]^2}\,dx$$

►

$$L=\int\sqrt{1+\left[\frac{d}{dx}\left(\frac{2\sqrt{2}}{3}x^{\frac{3}{2}}\right)\right]^2}\,dx$$

►

$$L=\int_0^2\sqrt{1+\left[\frac{d}{dx}\left(\frac{2\sqrt{2}}{3}x^{\frac{3}{2}}\right)\right]^2}\,dx$$

►

$$L=\int_0^2\sqrt{1+\left[\frac{d}{dx}\left(\frac{2\sqrt{2}}{3}x^{\frac{3}{2}}\right)\right]}\,dx$$

►

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(Marks: 1) - Please choose one

$\frac{2}{3}$ is known as

► An even number
► Irrational Number
► A natural Number

► Rational Number

(Marks: 1) - Please choose one

For a function *f*, let $f'(x_n)=0 \quad for\ some\ n$.

Does the Newton's Method works for approximating the solution of $f(x)=0$?

► Yes
► No

(Marks: 1) - Please choose one

The Mean Value Theorem states that "Let function *f* be differentiable on (a,b) and continuous on [a, b], then there exist at least one point c in (a,b) where ……. "

▶ $f'(c) = \frac{f(b) - f(a)}{b - a}$

$f(c) = \frac{f(b) - f(a)}{b - a}$

▶

$f(c) = \frac{f(a) - f(b)}{b - a}$

▶

$f'(c) = \frac{f(a) - f(b)}{b - a}$

▶

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(Marks: 1) - Please choose one

$$\frac{d}{dx}[F(x)] = f(x)$$

If there is some function *F* such that then any function of the form $F(x) + C$ is ---------- of *f*(*x*)

▶ Derivative
▶ Antiderivative
▶ Slope
▶ Maximum value

(Marks: 1) - Please choose one

If *f* and *g* are continues function on an interval [a, b] and $f(x) \geq g(x) \ \ for \ \ a \leq x \leq b$, then area is bounded by the lines parallel to:

▶ X -axis
▶ Y-axis
▶ Both X -axis and Y-axis

(Marks: 1) - Please choose one

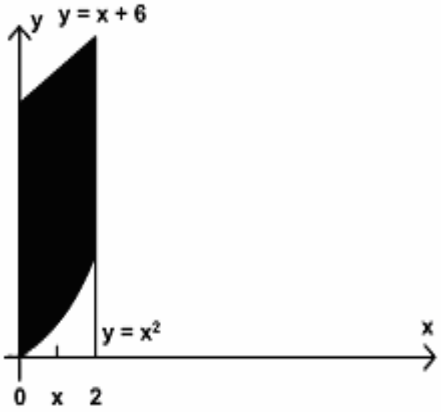
Sigma notation is represented by which of the following Greek letter?

▶ χ
▶ η

- Σ
- ψ

(Marks: 1) - Please choose one

In the following figure, the area enclosed is bounded below by **:**



- $y = x + 6$
- $y = x^2$
- $x = 2$
- $x = 0$

(Marks: 1) - Please choose one

Consider a function $h(x)$ and a constant c then

$\frac{d}{dx}\left((c)\{h(x)\}\right) =$ _____

- 0
- $\frac{d}{dx}\left(h(x)\right)$
- $\frac{d}{dx}\left(h(cx)\right)$
- $c\frac{d}{dx}\left(h(x)\right)$

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(Marks: 1) - Please choose one

Let the solid generated by the region enclosed between

$y = \sqrt{x}$; $x = 1, x = 4$ and the x-axis is revolved about the y-axis. Which of the following equation gives the volumes of a solid by cylindrical shells?

$$V = \int_1^4 2\pi x\sqrt{x}dx$$

▶

$$V = \int_1^4 2x\sqrt{x}dx$$

▶

$$V = \int_0^4 2x\sqrt{x}dx$$

▶

$$V = \int_{-4}^4 2x\sqrt{x}dx$$

▶

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(Marks: 1) - Please choose one

Let f is a smooth curve on the interval [a, b]. What is the arc length L of the curve f(x) defined over the interval [a, b]?

$$L = \lim_{\max \Delta x \to 0} \sum_{k=1}^{n} \sqrt{1 + (f'(x^*_k))}$$

▶

$$L = \sum_{k=1}^{n} \sqrt{1 + (f'(x^*_k))}\Delta x_k$$

▶

$$L = \lim_{\max \Delta x \to 0} \sum_{k=1}^{n} \sqrt{1 + (f'(x^*_k))^2}\Delta x_k$$

▶

$$L = \sum_{k=1}^{n} \sqrt{1 + (f(x^*_k))}\Delta x$$

▶

(Marks: 1) - Please choose one

Let $f(x)$ is a function such that as x approaches a real number a, either from left or right-hand-side, the function values increases or decreases unboundedly then

$\lim_{x \to a} f(x)$

- ► Exist
- ► Does not exist

(Marks: 1) - Please choose one

$\frac{d(\sec x)}{dx} =$

- ► $(\sec x)(\tan x)$
- ► $(\sec x)(\tan x)$
- ► $(co\sec x)(co\tan x)$
- ► $(co\sec x)(\tan x)$

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(Marks: 1) - Please choose one

At what points the two curves: $y = x^2 \ and \ y = x + 6$ intersect ?

- ► $x = 0 \ and \ x = 2$
- ► $x = 0 \ and \ x = 3$
- ► $x = 2 \ and \ x = 3$
- ► $x = -2 \ and \ x = 3$

(Marks: 1) - Please choose one

If f is continuous function such that $\lim_{x \to -\infty} f(x) = +\infty \ \ and \ \lim_{x \to +\infty} f(x) = +\infty$ then f has _____ on $(-\infty, +\infty)$

- ► maximum value but no minimum
- ► minimum value but no maximum
- ► both maximum and minimum value

For a graph to be symmetric about y-axis means, for each point (x,y) on the graph,the point ---------- is also on the graph

► (x , -y)
► (-x , y)
► (-x , -y)

(Marks: 1) - Please choose one

The graph $x = y^2$ is symmetric about ---------axis

► X-axis
► Y-axis
► Origin

(Marks: 1) - Please choose one

For a sequence $\{a_n\}$ if the ratio of successive terms $\frac{a_{n+1}}{a_n} \geq 1$ then the sequence is known as **:**

► Increasing
► Decreasing
► Nondecreasing
► Nonincreasing

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(Marks: 1) - Please choose one

$$a_n = \left\{\frac{1}{n}\right\}_{n=1}^{\infty}$$

Which of the following option is true for the sequence ?

► Increasing
► Decreasing
► Nonincreasing
► Nondecreasing

(Marks: 1) - Please choose one

If the partial sum of a series is finite then the series will/will be**:**

► Divergent
► Convergent
► Give no information

(Marks: 1) - Please choose one

If the geometric series $a + ar + ar^2 + ar^3 + ... + ar^{k-1} + ... \ \ where\ (a \neq 0)$,
$|r| < 1$
then which of the following is true for the given series?

► Converges
► Diverges
► Gives no information

(Marks: 1) - Please choose one

$\rho = \lim_{k \to +\infty} \frac{u_{k+1}}{u_k}$

If where $\rho > 1$ then the series $\sum u_k$ with positive terms will /will be.....?

► Convergent
► Divergent
► Give no information

(Marks: 1) - Please choose one

If a quantity y depends on another quantity x in such a way that each value of x determines exactly one value of y , we say that y is ..... of x

► relation
► function
► not a function
► not a relation

(Marks: 1) - Please choose one

$\frac{(x^2 - 4)}{(x - 2)}$

Domain of the function y = is

► $(-\infty, 2) U (2, +\infty)$
► $(-\infty, 2)$

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▶ $(-\infty, 0)$

(Marks: 1) - Please choose one

Tan(x) is continuous every where except at points

▶ $\pm\frac{k\pi}{2}(k=1,3,5,...)$

▶ $\pm\frac{k\pi}{2}(k=2,4,6,...)$

▶ $\pm\frac{k\pi}{2}(k=1,2,3,4,5,6,...)$

(Marks: 1) - Please choose one

$\lim_{x\to 0}\frac{\sin x}{x}$

= ----------

▶ -1
▶ 2
▶ 0
▶ 1

(Marks: 1) - Please choose one

How the series $1-3+5-7+9-11$ can be expressed in sigma notation?

▶ $\sum_{k=0}^{k=5}(-1)^k(2k+1)$

▶ $\sum_{k=1}^{k=5}(-1)^k(2k+1)$

▶

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$$\sum_{k=1}^{k=5}(2k+1)$$

▶

$$\sum_{k=1}^{k=5}(2k+1)$$

▶

(Marks: 1) - Please choose one

What is the sum of following series?

$1^3+2^3+3^3+4^3+_____+n^3$

$$\frac{n(2n)(2n+1)}{6}$$

▶

$$\frac{(n+1)(n+2)}{2}$$

▶

$$\left[\frac{n(n+2)}{2}\right]^2$$

▶

$$\left[\frac{n(n+1)}{2}\right]^2$$

▶

(Marks: 1) - Please choose one

$\frac{5}{7}\times1^2+\frac{5}{7}\times2^2+\frac{5}{7}\times3^2+\frac{5}{7}\times4^2....+\frac{5}{7}\times n^2=$ _____

$$\frac{5n(n+1)(2n+1)}{42}$$

▶

$$\frac{5n(n+1)}{14}$$

▶

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$\frac{5n^2(n+1)^2}{14}$

▶

$\frac{5(n+1)(2n+1)}{42}$

▶

(Marks: 1) - Please choose one

$\int_a^a f(x)dx =$ _____

If point a is in the domain of function f, then

▶ $f'(x)$

▶ $f(x)$

▶ 0

▶ 1

(Marks: 1) - Please choose one

If $a_1 > a_2 > a_3 > ..... > a_n > ....$, then a sequence $\{a_n\}$ is .....

▶ Increasing

▶ Nondecreasing

▶ Decreasing

▶ Nonincreasing

(Marks: 1) - Please choose one

$$\sum_{k=1}^{\infty}(-1)^n a_k$$

A series of the form is called _____.

▶ Alternating series

▶ Geometric series

▶ Arithmetic series

▶ Harmonic series

(Marks: 1) - Please choose one

Which of the following is the Maclaurin series for e^x ?

$$1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\ldots+\frac{x^k}{k!}+\ldots$$

▶

$$x+\frac{x^2}{2!}+\frac{x^3}{3!}+\ldots+\frac{x^k}{k!}+\ldots$$

▶

$$1+x+\frac{x^3}{3!}+\ldots+\frac{x^k}{k!}+\ldots$$

▶

$$1-x+\frac{x^3}{3!}-\ldots-\frac{x^k}{k!}-\ldots$$

▶

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(Marks: 1) - Please choose one

Which of the following is the work done *W* if an object moves in the positive direction along a coordinate line while subject to a force *F(x)* in the direction of motion over an interval [0,3]?

$$W=\int_{2}^{3}3xdx$$

▶

$$W=\int_{0}^{3}3xdx$$

▶

$$W=\int_{0}^{3}F(x)dx$$

▶

$$W=\int_{3}^{0}F(x)dx$$

▶

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(Marks: 1) - Please choose one

Which of the following is the spring constant *k* if a spring whose natural length is 2*m* exerts a force of 3*N* when stretched 1*m* beyond its natural length?

▶ 3 *x*

▶ 3 *N/m*

▶ 2 *m*

► 3 *m/N*

(Marks: 2)

Find the limits of the integral indicating the area bounded by the curves $y = x^2 \; and \; y = x + 6$.

(Marks: 2)

What will be the amount of work done if an object moves 7*m* in the direction of a force of 70*N*?

(Marks: 2)

Evaluate the following integral by substitution method.

$$\int x\,(2x^2+1)^{\frac{2}{3}}\,dx$$

(Marks: 3)

Evaluate the following integral:

$$\int \frac{5-6\sin^2 x}{\sin^2 x}\,dx$$

(Marks: 3)

Find the spring constant '*k*'; if a force of 10*N* is required to stretch a spring from its natural length of 4.8*m* to a length of 6.8*m*?

(Marks: 3)

Find a definite integral indicating the area of the surface generated by revolving the curve $y = \sqrt[3]{3x} \;\; ; \;\; 0 \le y \le 4$ about the $x-$ axis. But do not evaluate the integral.

(Marks: 5)

Let $\frac{d}{dx}\left[f(x)\right] = 12x^2 - 6x + 1$. Find $f(x)$

(Marks: 5)

Determine whether the sequence $\{a_n\}$ converges or diverges; if it converges then find its limit;

$$a_n = \frac{3n^4+1}{4n^2-1}$$

where

(Marks: 5)

Use the cylindrical shell to find the volume of the solid generated when the region enclosed by the curve $y = x^3, x = 1, y = 0$ is revolved about the *y*-axis.

(Marks: 10)

Find the area of the region that is enclosed by the curves $y = x^2$ and $y = \sqrt{x}$ between $x = \frac{1}{4}$ and $x = 1$.

This paper is solved by our best knowledge. In the case of any error/correction/suggestion, please contact at gulshanvu@yahoo.com, with reference to the concerned paper's number.

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FINALTERM EXAMINATION
Fall 2008
(Session - 1)

Calculus & Analytical Geometry-I

Question No: 1 (Marks: 1) - Please choose one

If $y = f(x)$ then the average rate of change of y with respect to x over the interval $[x_0, x_1]$ is the …… Joining the points $(x_0, f(x_0)) and (x_1, f(x_1))$ on the graph of f

- ► Slope of the secant line
- ► Slope of tangent line
- ► Secant line
- ► none of these

Question No: 2 (Marks: 1) - Please choose one

Natural domain of $\frac{(x^2-4)}{(x-2)}$ is

- ► $(-\infty, 2) U (2, +\infty)$
- ► $(-\infty, 2)$
- ► $(-\infty, 0)$
- ► None of these

Question No: 3 (Marks: 1) - Please choose one

The equation $(x+4)^2 + (y-1)^2 = 6$ represents a circle having center at …….. and radius .....

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► $(-4,1),\sqrt{6}$

► $(-4,1),6$

► $(-4,-1),\sqrt{6}$

► None of these

Question No: 4 (Marks: 1) - Please choose one

The series $\sum u_k$ be a series with positive terms and suppose that $\rho = \lim_{k\to\infty} \sqrt[k]{u_k} = \lim_{k\to\infty} (u_k)^{\frac{1}{k}}$ if $\rho > 1$ then the series

► Converges

► **Diverges**

► May converge or diverge

► None of these

Question No: 5 (Marks: 1) - Please choose one

The series $\sum u_k \text{ and } \sum v_k$ are convergent series then $(\sum u_k + \sum v_k) \text{ and } (\sum u_k - \sum v_k)$ will be ……and…….

► Convergent, convergent

► Divergent, divergent

► **Convergent, divergent**

► Divergent, convergent

Question No: 6 (Marks: 1) - Please choose one

The notation $\{\frac{1}{2^n}\}_{-1}^{n}$ represents the sequence

► $2,1,\frac{1}{2},\frac{1}{4},...$

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- ► $0,1,2,3...$
- ► $0,1,\frac{1}{2},\frac{1}{4},...$
- ►
- ► None of these

Question No: 7 (Marks: 1) - Please choose one

If f is continuous on (a,b] but does not have a limit from the right then the integral

$$\int_a^b f(x)dx = \lim_{l \to a} \int_l^b f(x)dx$$

defined by is called …….. Integral

- ► Improper
- ► Proper
- ► None of these

Question No: 8 (Marks: 1) - Please choose one

An object is displaced 1m by a force of 1N then the work done W is

- ► 2
- ► 0
- ► None of these
- ► **1**

Question No: 9 (Marks: 1) - Please choose one

If f is a smooth function on [a,b] then the arc length L of the curve y=f(x) from x=a to x=b will be

- ► $L = \int_a^b \sqrt{1+[f'(x)]^2}\,dx$
- ► $L = \int_a^b \sqrt{1+[f'(x)]}dx$
- ►

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$$L=\int_0^a \sqrt{1+[f'(x)]}dy$$

►

► None of these

Question No: 10 (Marks: 1) - Please choose one

If f is a smooth function on [0,3] then the arc length L of the curve y=f(x) from x=0 to x=3 will be

$$L=\int_0^3 \sqrt{1+[f'(x)]^2}dx$$

►

$$L=\int_a^b \sqrt{1+[f'(x)]^2}$$

►

$$L=\int_0^3 \sqrt{1+[f'(x)]^2}dy$$

►

► None of these

Question No: 11 (Marks: 1) - Please choose one

By using cylindrical shell to find volume of the solid when the region R in the first quadrant enclosed between $y=3x \text{ and } y=2x^2$ is revolved about the x-axis

$$V=\int_0^{\frac{3}{2}} 2\pi x(3x-2x^2)dx$$

►

$$V=\int_0^{\frac{3}{2}} x(3x-2x^2)dx$$

►

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$$V = \int_0^{\frac{3}{2}} 2\pi(3x - 2x^2)dx$$

►

► None of these

Question No: 12 (Marks: 1) - Please choose one

By using cylindrical shell to find volume of the solid when the region R in the first quadrant enclosed between $y = x \text{ and } y = x^2$ is revolved about the y-axis is represented by

$$V = \int_0^3 2\pi x(x - x^2)dx$$

►

$$V = \int_0^1 x(x - x^2)dx$$

►

$$V = \int_0^1 2\pi(x - x^2)dx$$

►

► None of these

Question No: 13 (Marks: 1) - Please choose one

If

$$\int_a^a f(x)dx =$$

a is in the domain of f, then

► $f'(x)$

► $f(x)$

► **0**

► None of these

Question No: 14 (Marks: 1) - Please choose one

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$$\int_0^2 x^2 dx$$

Consider the integral , the area on right is bounded by

- ▶ $y = x^2$
- ▶ $x = 2$
- ▶ $x = 0$
- ▶ None of these

Question No: 15 (Marks: 1) - Please choose one

The series $1-3+5-7+9-11$ may written as in sigma notation

- ▶ $\sum_{k=0}^{k=5}(-1)^k(2k+1)$
- ▶ $\sum_{k=1}^{k=5}(-1)^k(2k+1)$
- ▶ $\sum_{k=1}^{k=5}(2k+1)$
- ▶
- ▶ None of these

Question No: 16 (Marks: 1) - Please choose one

$4^2+5^2+6^2+7^2$ in sigma notation may be represented as

- $\sum_{k=2}^{k=7}k^2$
- ▶
- $\sum_{k=2}^{k=7}(k+1)^2$
- ▶

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$$\sum_{k=4}^{k=7} k^2$$

►

► None of these

Question No: 17 (Marks: 1) - Please choose one

If a function f is …….. on a closed interval [a,b] ,then f has both a maximum and minimum value on [a,b]

► **Continuous**
► Discontinuous
► Differentiable
► None of these

Question No: 18 (Marks: 1) - Please choose one

Let f be a function on an interval, and $x_1 \text{ and } x_2$ denote the points in that interval, if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ then the we can say that f is

► Increasing function
► **Decreasing function**
► Constant function
► None of these

Question No: 19 (Marks: 1) - Please choose one

If a function satisfies the conditions

f(c) is defined

$\lim_{x \to c^+} f(x)$ Exists

$\lim_{x \to c^+} f(x) = f(c)$

Then the function is said to be

► **Continuous at c**

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- ► Continuous from left at c
- ► Continuous from right at c
- ► None of these

Question No: 20 (Marks: 1) - Please choose one

For a function f(x) to be continuous on interval (a,b) the function must be continuous

- ► **At all point in (a,b)**
- ► Only at a and b
- ► At mid point of a and b
- ► None of these

Question No: 21 (Marks: 2)

$$a_{n+1} = \frac{1}{3}(a_n + \frac{1}{a_n}) \; for \; n \geq 1 \; and \; a_1 = 2$$

Write down the first two term of the sequence

Question No: 22 (Marks: 2)

Find the integral of the surface area of the portion of the sphere generated by revolving the curve $y = \sqrt{2 - x^2}, 0 \leq x \leq \frac{1}{3}$

(Note: Just find the integral do not solve the integral)

Question No: 23 (Marks: 2)

$\int_2^5 f(x)dx$ $\int_2^3 f(x)dx = 7, \int_3^4 f(x)dx = 2, \int_5^4 f(x)dx = 5$

Calculate if

$\int_2^3 f(x)dx = 7, \int_3^4 f(x)dx = 2, \int_5^4 f(x)dx = 5$

$\int_2^5 f(x)dx = 7 + 2 - 5 = 4$

Question No: 24 (Marks: 3)

Find

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the first two Taylor polynomials for $\ln x$ about $x = 3$

Question No: 25 (Marks: 3)

Let the curve $y = x^{\frac{3}{2}} \;;\quad 0 \le y \le 2$, then find the surface area generated by revolving the curve. (But do not evaluate)

Question No: 26 (Marks: 3)

Express the sum $\frac{1}{1}+\frac{1}{4}+\frac{1}{9}+\ldots\ldots+\frac{1}{7225}$ in sigma notation but do not evaluate.

$\frac{1}{1}+\frac{1}{4}+\frac{1}{9}+\ldots\ldots+\frac{1}{7225}$

$\sum_{k=1}^{7225} k^3+1$

Question No: 27 (Marks: 5)

Find the first four nonzero terms of the Taylor series generated by f at $x = a$

$f(x) = \frac{1}{1-x} \quad at \quad x = 2$

Question No: 28 (Marks: 5)

Evaluate the Definite Integral using the First fundamental Theorem of Calculus

$\int_0^1 (x^5 - x^3 + 2x)dx$

$Let\ u = (x^5 - x^3 + 2x)$

$\int_0^1 (u)dx$

Question No: 29 (Marks: 5)

Express the definite integrals as limits (Do not evaluate the integrals)

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$$\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} (1+Cosx)\,dx$$

$$\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} (1+Cosx)\,dx$$

$$\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \lim_{\max xk \to 0} \sum_{\frac{-\pi}{2}}^{\frac{\pi}{2}} (1+\cos x)dx$$

Question No: 30 (Marks: 10)

Find the region enclosed by the curves and also find the area

$y = x^2$, $y = \sqrt{x}$, $x = \frac{1}{4}$, $x = 1$

Question No: 31 (Marks: 10)

Use x^*_k as the left end point of each subinterval to find the area under

$y = mx \;\; over\, the \,\text{int}\, erval\,[a,b], where\, m > 0\, and\, a \geq 0$

Solution on next page

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Suppose $a=1\ b=2\ so[a,b]=[1,2]$

$x_k^* = x_{k-1} = a+(k-1)\Delta x \quad (formula\ for\ left\ end\ point)$

$$\Delta x = \frac{b-a}{n} = \frac{2-1}{n} = \frac{1}{n}$$

Suppose kth has area

$f(x_{k^*)}\Delta x = x_{k^*}\Delta x$

$$\left[1+\frac{k}{n}\right]\Delta x$$

$$\left[1+\frac{k}{n}\right]\frac{1}{n}$$

$$\sum_{k=1}^{n} f(x_{k^*})\Delta x = \sum_{k=1}^{n}[1+k-1]$$

Area by solving

$$A = \lim_{\delta x\to 0}\sum_{k=1}^{n} f(x_{k^*})\Delta x = \lim_{\delta x\to 0}\left[\frac{3}{2}-1+\frac{1}{2n}\right]$$

$$= \frac{3}{2}-1+0$$

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